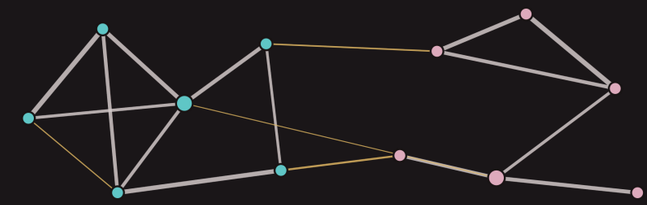
Co-expression Networks: Construction and Analysis



Tutorial VT1 | ISCB-Latin America SoIBio
SPBBC Conference on Bioinformatics 2026

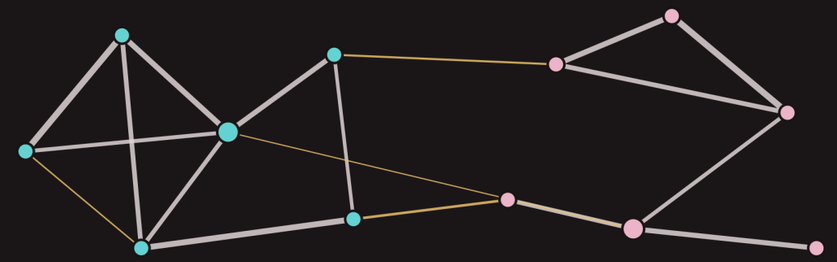
Dr. Deisy Morselli Gysi | Federal University of Paraná (UFPR), Brazil

2026-11-09

Today's Scope

Lithium has treated **bipolar disorder** for more than 70 years, and we still do not fully understand what it does.

Our question: in the blood of patients on and off lithium, which genes vary together, how are those relationships organized, and which changes belong to the **disease** and which to the **drug**?

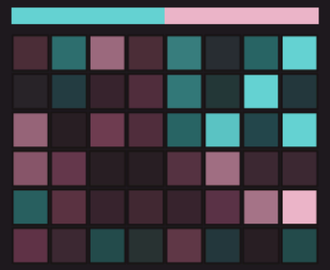


ISCB-Latin America SoIBio SPBBC Conference on Bioinformatics 2026, organized by the Sociedad Peruana de Bioinformática y Biología Computacional (SPBBC), the Iberoamerican Society of Bioinformatics (SoIBio), and the International Society for Computational Biology (ISCB).

The Course at a Glance

01 DATA

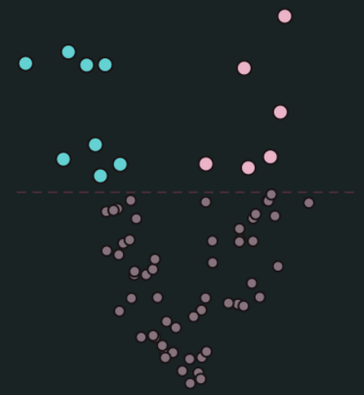
What was measured?



GSE124326 · QC

02 DE BASELINE

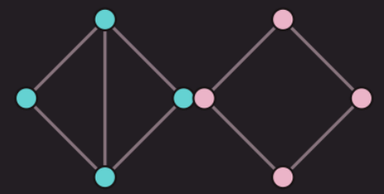
Which genes shift?



limma

03 NETWORKS

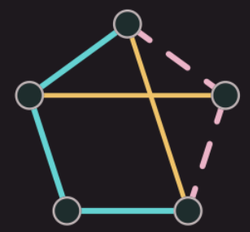
Which genes move together?



WGCNA · ARACNE · wTO

04 COMPARE

What is rewired?



Consensus · CoDiNA

05 FUNCTION

What does it mean?



ORA · GSEA

> We keep one dataset and one gene set from start to finish, so each step builds on the last: **from measurements, to relationships, to differences, to biology**.

Learning Outcomes • The Route

By the end of this block, you can:

1

Graphs

Read a network: degree, hubs, neighbourhoods, and modules.

2

Data

Prepare GSE124326, remove technical effects, choose the network genes, and set a quick limma baseline.

3

Networks

Build and compare WGCNA, ARACNE, and wTO networks.

4

Compare

Find robust links (consensus) and rewired links (CoDiNA).

5

Function

Interpret modules and gene classes with ORA and GSEA.

6

Hands-on

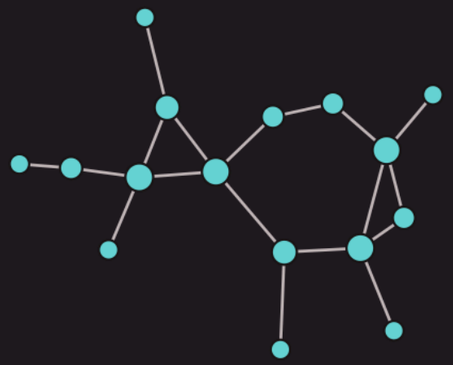
Practise every step in the companion notebook, and know each step's limits.

Part I · Why Networks?

Graph theory gives us the language; biology gives us the questions.

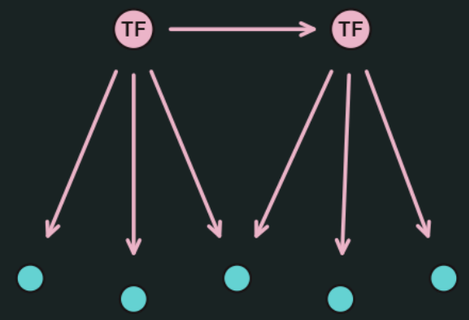
Biological Networks Come in Many Flavors

01 PPI



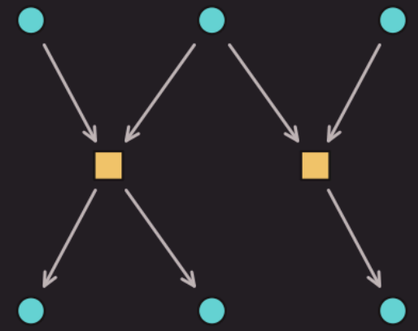
Physical binding

02 REGULATORY



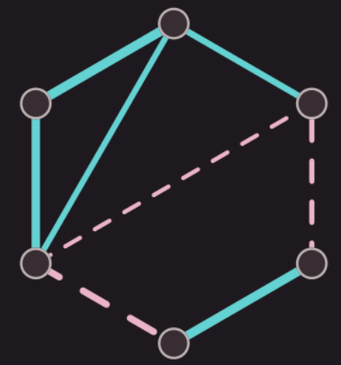
TF → target

03 METABOLIC



Metabolite · reaction

04 CO-EXPRESSION



Correlated expression

Network	Nodes	Links mean	Directed?	Typical source
PPI	Proteins	Physical binding	No	Yeast two-hybrid, AP-MS
Regulatory	TFs and targets	Regulation	**Yes**	ChIP-seq, perturbation
Metabolic	Metabolites, reactions	Conversion	Yes	Genome-scale models
Co-expression	Genes	**Similar expression**	No	**Transcriptomes**

Graphs and Networks

Graph · the mathematical object

- $G = (V, E)$: a set of vertices V and edges $E \subseteq V \times V$
- $n = |V|$ vertices, $m = |E|$ edges
- Defined only by **who connects to whom**
- Language: vertices and edges

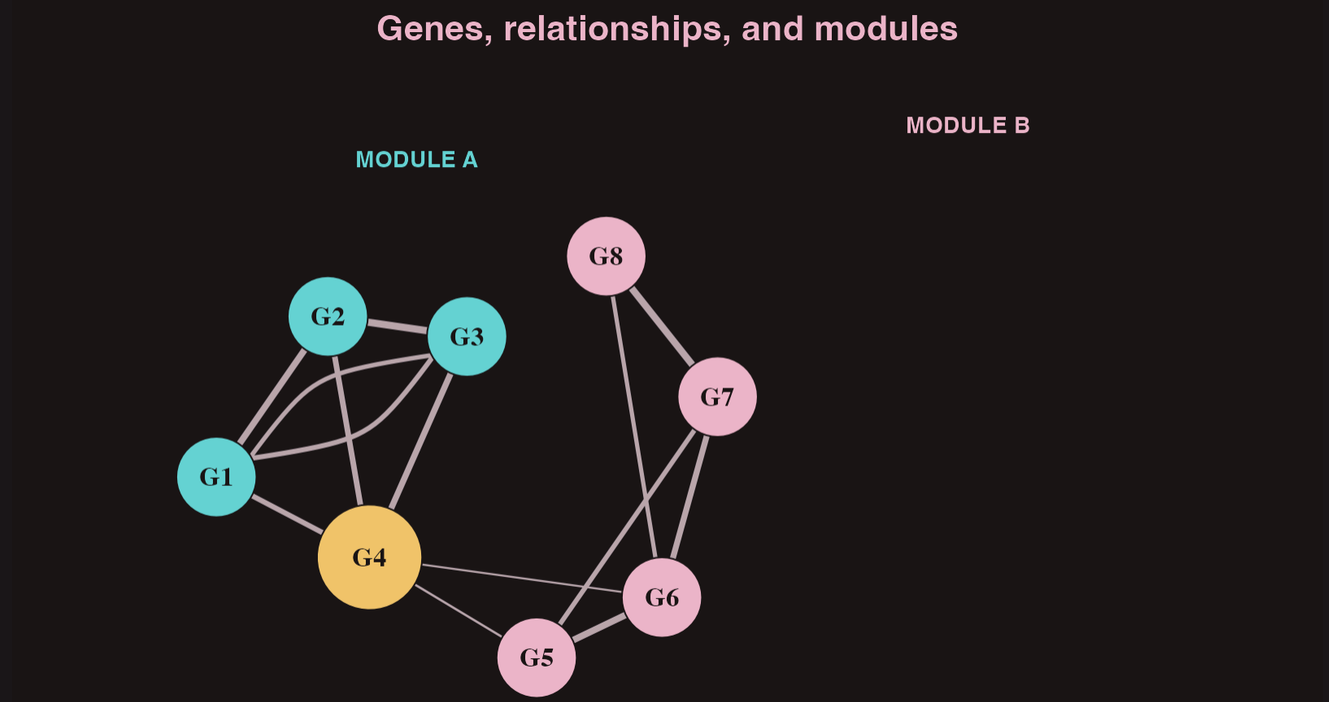
Network · the real system

- A graph whose nodes and links **carry meaning and attributes**
- Genes, proteins, metabolites, patients
- Focus on organization, dynamics, and function
- Language: nodes and links

> A co-expression network is a **graph built from data**: its nodes are genes, and its links summarize how similarly they vary across samples. Every graph concept we use next applies to it directly.

What Is a Biological Network?

In a gene network, the nodes are genes and the links join genes that behave alike. Modules and hubs are not put in by hand: they emerge from the pattern of links.



NODES are genes.

LINKS connect gene pairs; co-expression links are undirected.

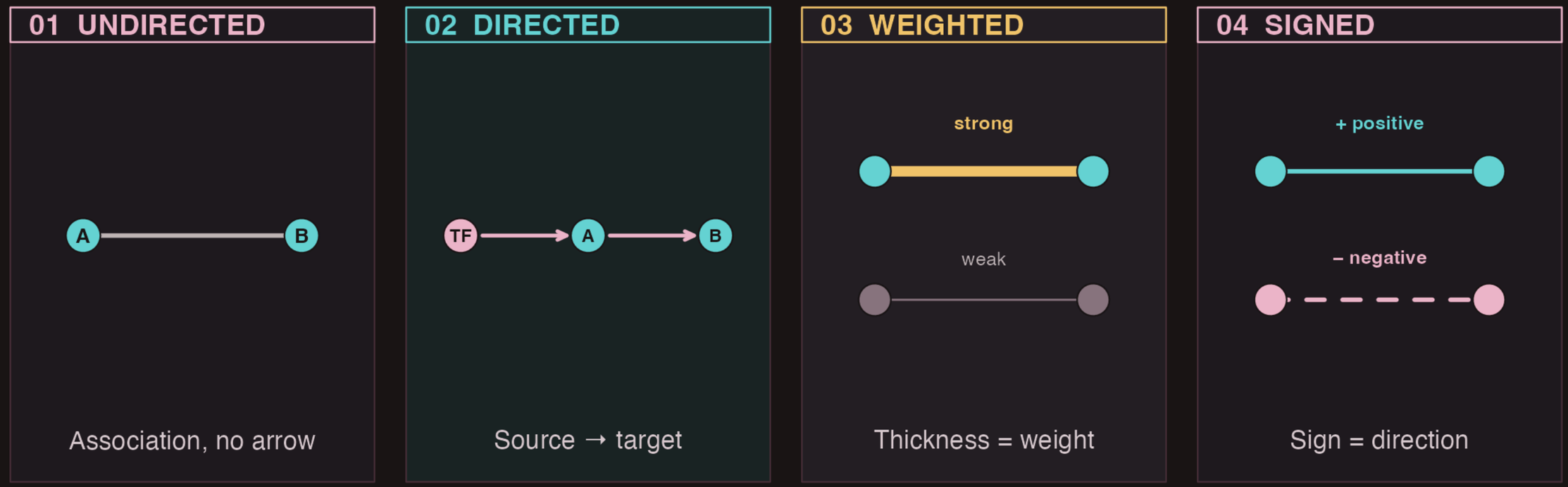
WEIGHT encodes strength; thicker means stronger.

MODULES are dense gene neighborhoods.

HUBS have many links; **G4** bridges modules.

Graphs Can Encode Different Relationships

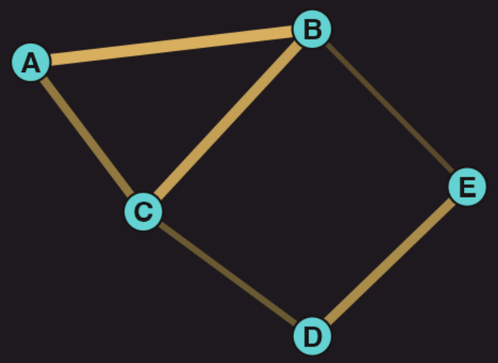
The same nodes can be joined in different ways. An edge can have a direction, a weight for its strength, and a sign that separates positive from negative relationships.



Co-expression networks are typically **undirected, weighted, and signed**. They do not encode causal direction.

Representing a Network

01 GRAPH



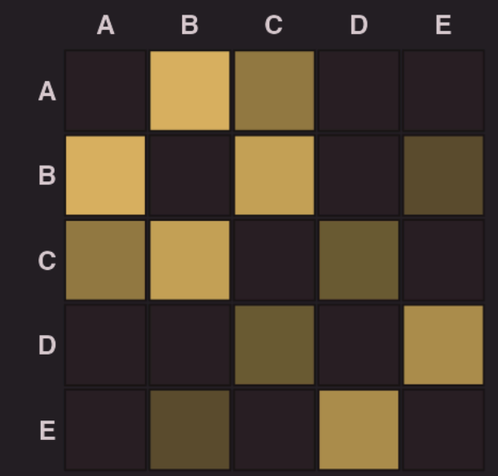
5 genes, 6 weighted links

02 EDGE LIST

from	to	w
A	B	0.9
A	C	0.6
B	C	0.8
C	D	0.4
D	E	0.7
B	E	0.3

from · to · weight

03 ADJACENCY MATRIX



A[ij] = w[ij], symmetric

A **correlation matrix is already a dense, weighted adjacency matrix**. Each co-expression method starts from a matrix of this kind. They differ in how they score each pair and which entries they keep.

Basic Graph Vocabulary

Concept	Definition	In a gene network
Neighbor	A node joined by an edge	A gene associated with the gene of interest
Degree $k_i = \sum_j A_{ij}$	Number of edges at node i	Number of gene associations
Strength $s_i = \sum_j \lvert w_{ij} \rvert$	Sum of edge weights at node i	Total weighted association (“connectivity” in WGCNA)
Path	A sequence of connected edges	A route through associated genes
Hub	Node with unusually many or strong links	Candidate for prioritization, not automatically a regulator
Module	Densely interconnected group	Candidate coordinated gene program

These words describe structure only. To say what a hub or a module *does*, we need annotation and, eventually, experiments.

Degree Distributions: Random vs Scale-free

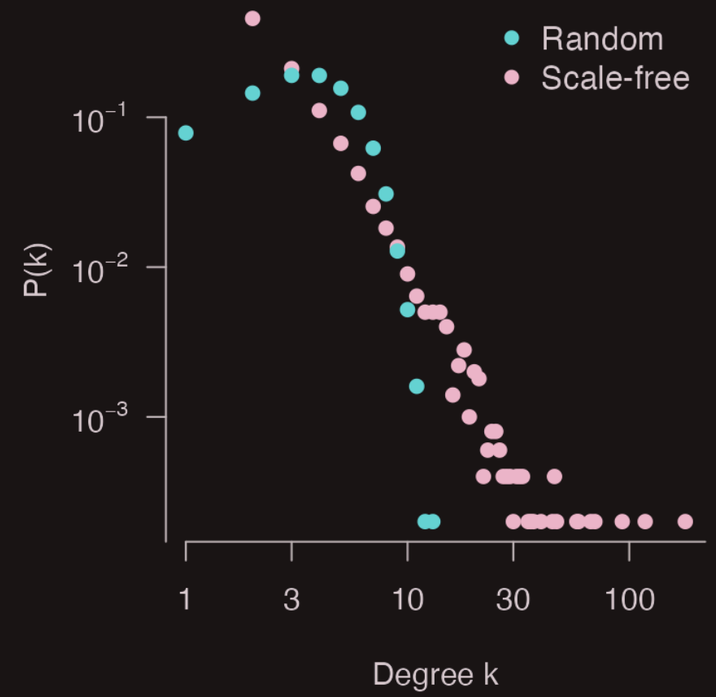
Random (Erdős–Rényi)



Scale-free (Barabási–Albert)



Degree distribution (log–log)



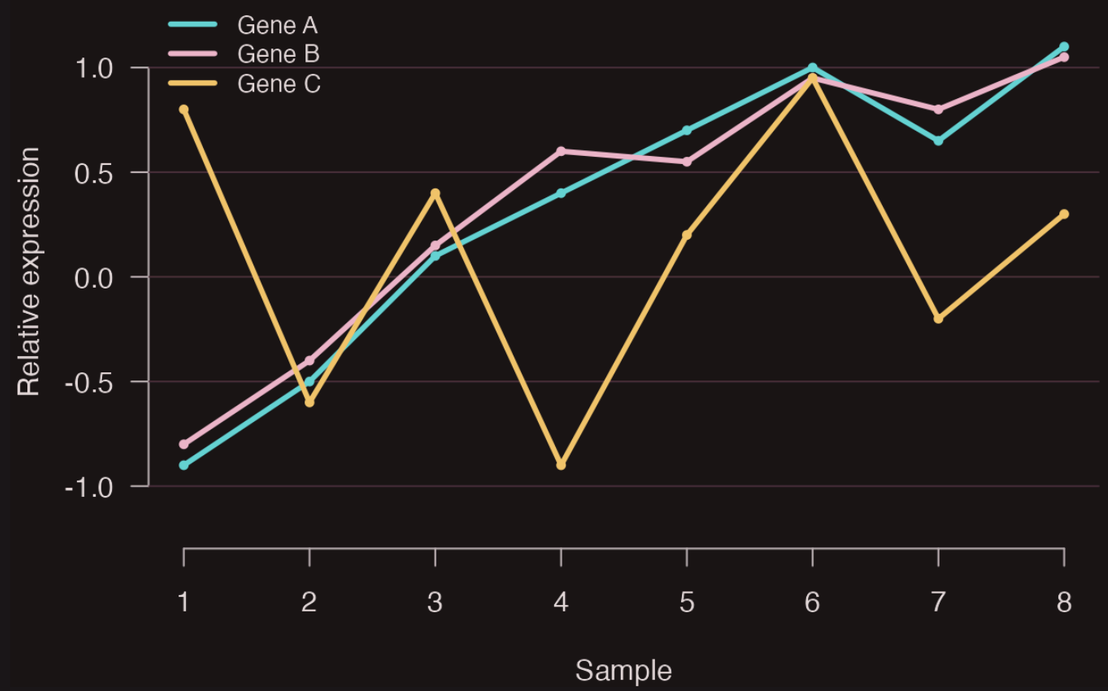
Random: degrees cluster around $\langle k \rangle = 2m/n$; hubs are vanishingly rare.

Scale-free: $P(k) \sim k^{-\gamma}$, typically $2 < \gamma < 3$; a few **hubs** hold many links. **WGCNA uses this shape to choose its soft threshold.**

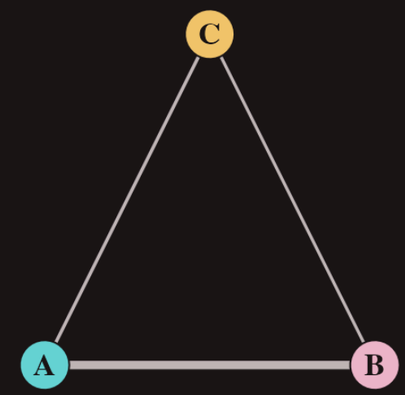
Why Does Co-expression Matter?

Genes rarely act alone. When two genes rise and fall together across samples, we draw a strong link between them; when their profiles are unrelated, the link is weak.

Expression profiles across samples



Weighted co-expression network



Simulated profiles, for illustration.

Co-expression Is a Pattern, Not a Mechanism

Genes are connected when their expression profiles are statistically similar across samples.

Useful for:

- organizing coordinated gene programs
- prioritizing modules and candidate hubs
- comparing network organization between conditions

Not established by correlation alone:

- direct physical interaction
- causal regulation
- direction of influence

> A link tells us that two genes vary together. *Why* they do is a separate question, and usually an experimental one.

Part II · Data and Preparation

GSE124326 · whole blood · 112 controls · 226 bipolar I patients, on and off lithium

Case Study · Lithium in Bipolar Disorder

Whole-blood RNA-seq from **GSE124326** (Krebs *et al.*, 2020), as kept in the authors' final analysis: **112 controls** and **226 patients with bipolar I disorder**, **152 taking lithium** and **74 not taking it**. The original study found a strong lithium effect on expression; we ask whether lithium also changes **how genes vary together**, and whether we can tell that apart from the disease.

Input field	Meaning
`expression`	$\log_2$-CPM, genes × samples (HGNC symbols)
`treatment`	`Control`, `no Lithium` or `Lithium`
`sex`, `plate`	Covariates: sex and sequencing plate
`neutrophils`, …	Estimated blood-cell fractions

CPM = counts per million: a gene's reads ÷ the sample's total reads × 10^6, so samples sequenced to different depths become comparable. $\log_2$(CPM) makes the values roughly symmetric. `R/prepare_gse124326.R` builds it from the GEO counts.

> Lithium was **not randomized**: differences can reflect lithium, illness course, or co-medication, and lithium **raises neutrophil counts**. Controls were sequenced mostly on plates 3–5, so plate goes into every model.

Start with the Matrix Contract

Most network packages expect **samples in rows and genes in columns**. Keep the sample metadata aligned to the expression matrix before fitting any model.

	Gene A	Gene B	Gene C
Sample 1	4.1	7.2	2.8
Sample 2	4.5	6.9	3.2
Sample 3	3.9	7.5	2.6

```r
input_file <- "data/gse124326_bd_lithium.rds"
stopifnot(file.exists(input_file))
course_data <- readRDS(input_file)

expression_matrix <- course_data$expression   # genes x samples
sample_info <- course_data$sample_info
stopifnot(identical(colnames(expression_matrix), sample_info$sample))
stopifnot(!anyNA(expression_matrix))
```

Inspect the Matrix and the Design

```r
dim(expression_matrix)
```

```
[1] 13013   338
```

```r
expression_matrix[1:3, 1:4]
```

```
         11R0539 11R0540 11R0541 11R0545
A1BG-AS1   2.020   1.932   1.803   0.640
A2M       -0.193   1.021   0.497  -0.241
A2M-AS1    4.890   5.293   5.499   5.755
```

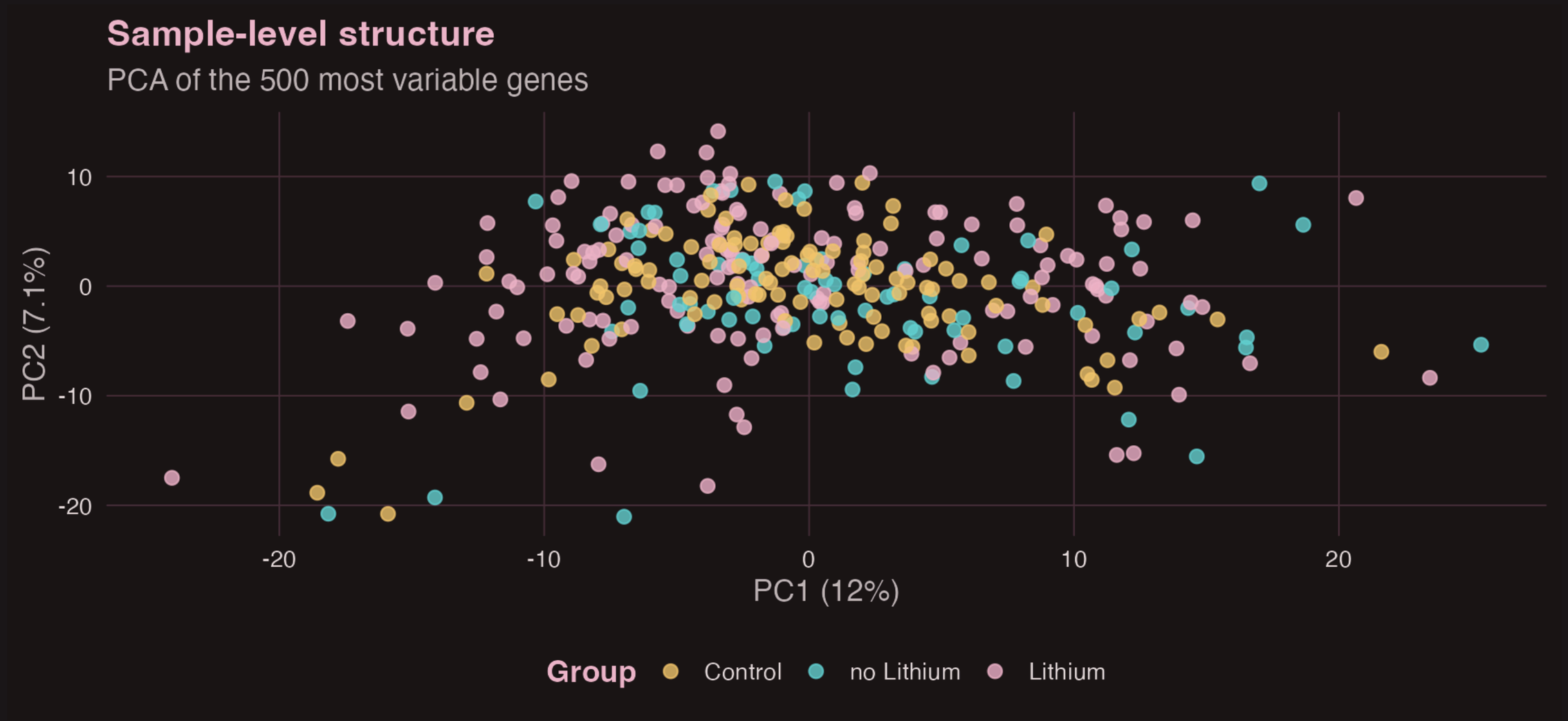
```r
table(sample_info$treatment, sample_info$sex)
```

```
            F  M
Control    67 45
Lithium    90 62
no Lithium 35 39
```

Check the orientation: rows are genes here; transpose only at the package boundary that expects samples × genes. Sex is unbalanced between groups, which is one reason it goes into the model.

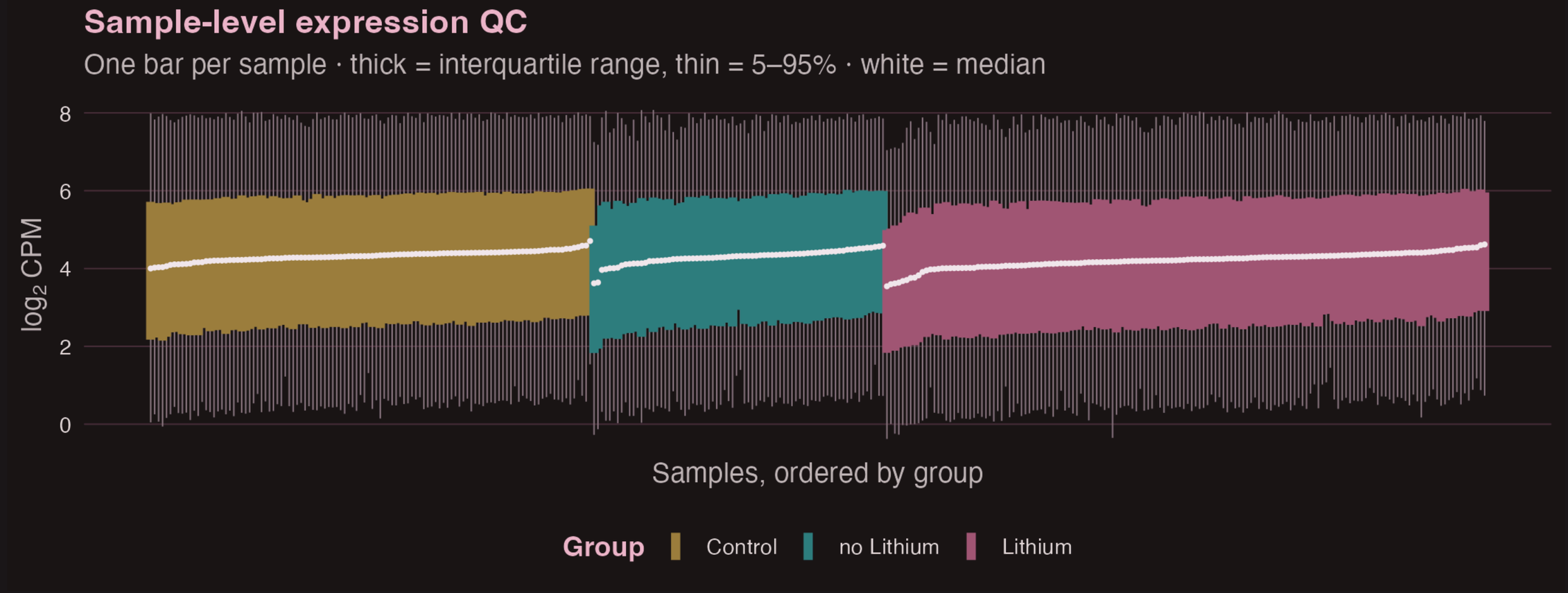


Quality Checks Before Inference



Look for sample outliers, dominant batches, and unexpected group separation. A PCA plot is a diagnostic, not a pass/fail test.

Check Expression Distributions by Sample



Medians line up because CPM already corrects for sequencing depth. A sample whose box sits apart deserves a closer look before network inference.

Essential Preprocessing Decisions

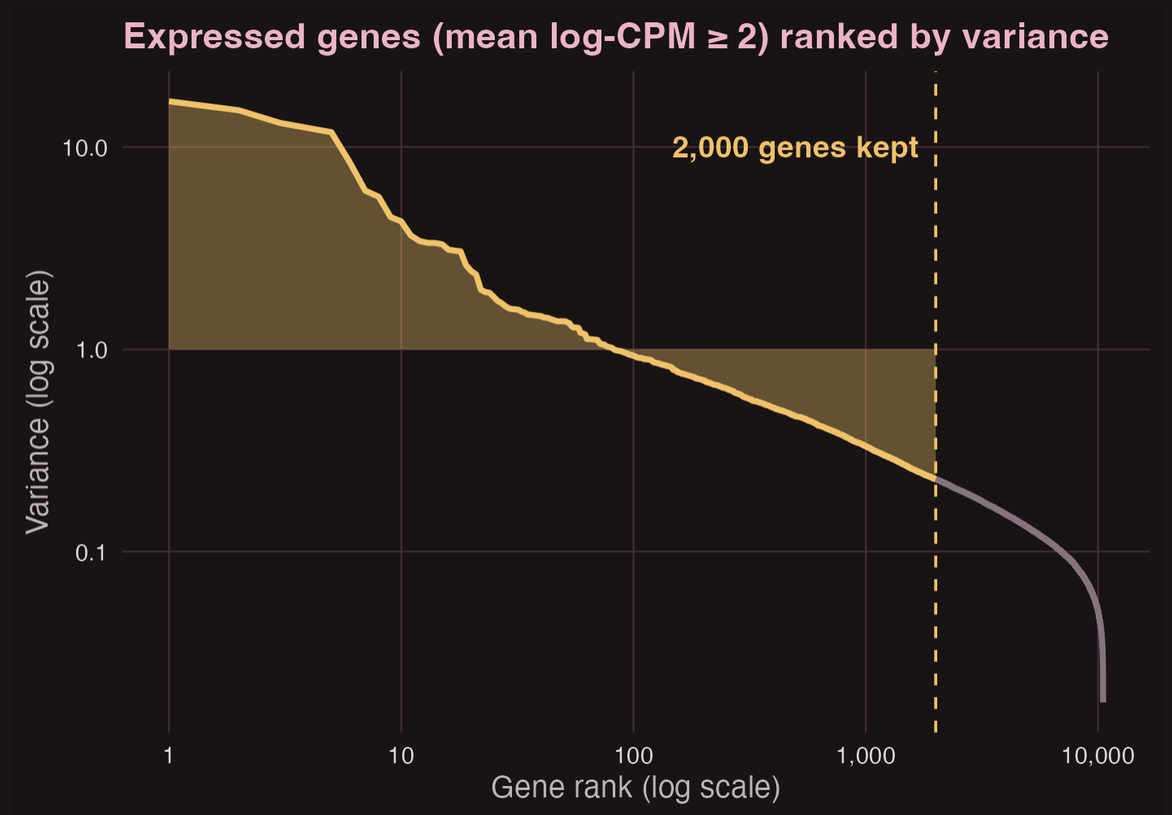
Count data need a transformation before correlation-based methods. Which one?

Transform	Package	Use it when
$\log_2$-CPM	edgeR / limma	Simple and transparent; fine with decent depth once low counts are filtered
`voom`	limma	DE with limma: log-CPM plus a precision weight per observation
`vst`	DESeq2	Networks, PCA, clustering with **many** samples: fast, stabilizes variance
`rlog`	DESeq2	**Few** samples (≲ 30) with very different library sizes; slow for large studies

Also: filter lowly expressed genes, check outliers and batches, and keep labels aligned with the matrix. **Here:** $\log_2$-CPM (338 samples, filtered); sex and plate go into limma and are regressed out before the networks.

> The quality of the final network depends more on preprocessing than on the choice of algorithm.

Select the 2,000 Most Variable Genes



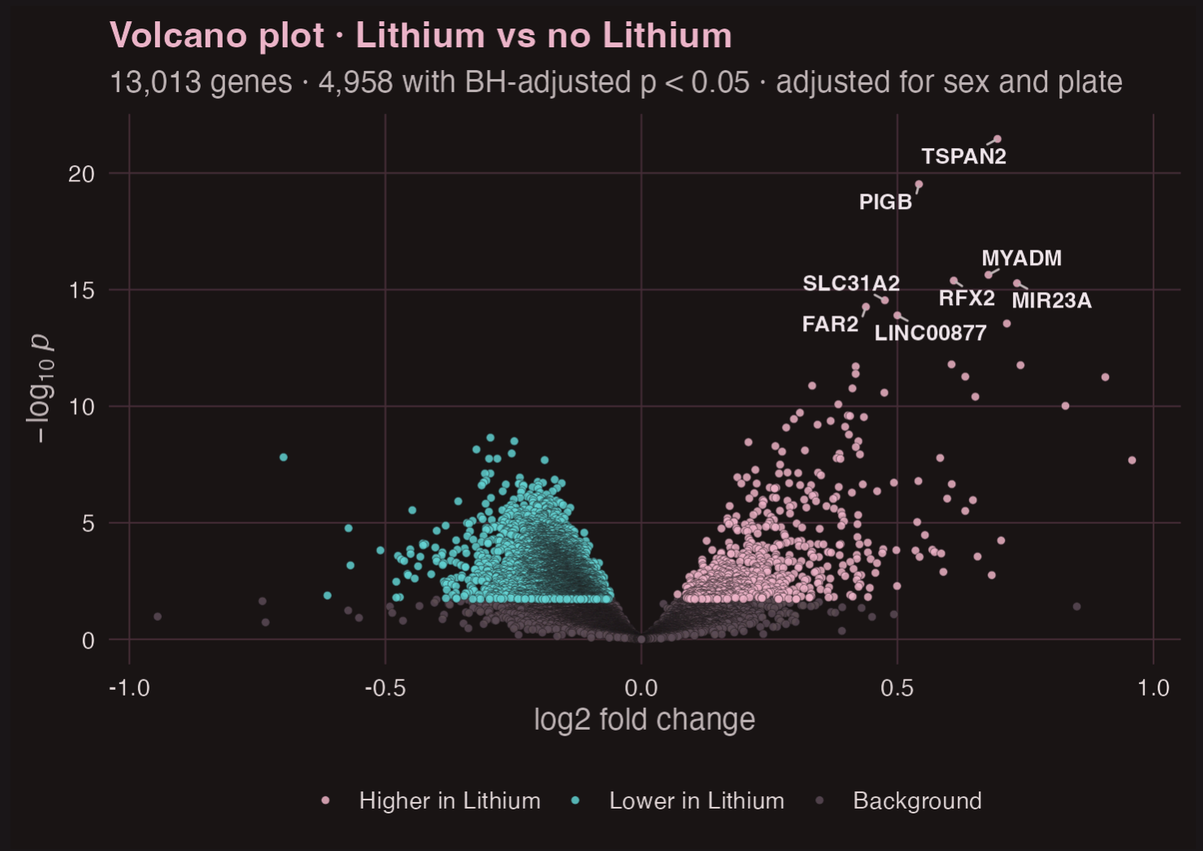

```r
# remove plate and sex, keep the group effects
adjusted_matrix <- limma::removeBatchEffect(
  expression_matrix, batch = sample_info$plate,
  covariates = as.numeric(sample_info$sex == "M"),
  design = treatment_design)
# expressed genes only: low counts are noisy
expressed <- rowMeans(expression_matrix) >= 2
gene_variance <- apply(adjusted_matrix[expressed, ], 1,
 var)
selected_genes <- names(sort(gene_variance,
                  decreasing = TRUE))[1:2000]
```

Why adjust first? Otherwise sex genes (XIST) and plate effects dominate the variance and the network.

Why expressed, then variance? At low counts, log values vary mostly by sampling noise; genes that barely vary add noise correlations.

Why not DEGs? Choosing genes for differing between groups would build the module–trait association into the selection. A variance filter **ignores the labels**.

A Quick Baseline · Differential Expression



Before building networks, we run the standard analysis once, as a **baseline** to compare against at the end.

One limma model on all expressed genes, adjusted for sex and plate, with two contrasts: **lithium** (on vs off, within bipolar disorder) and **diagnosis** (patients off lithium vs controls).

Lithium moves thousands of genes, each by a little. But DE only sees **changes in average level**. Networks ask whether genes **vary together**, and that is where we spend the rest of the session.

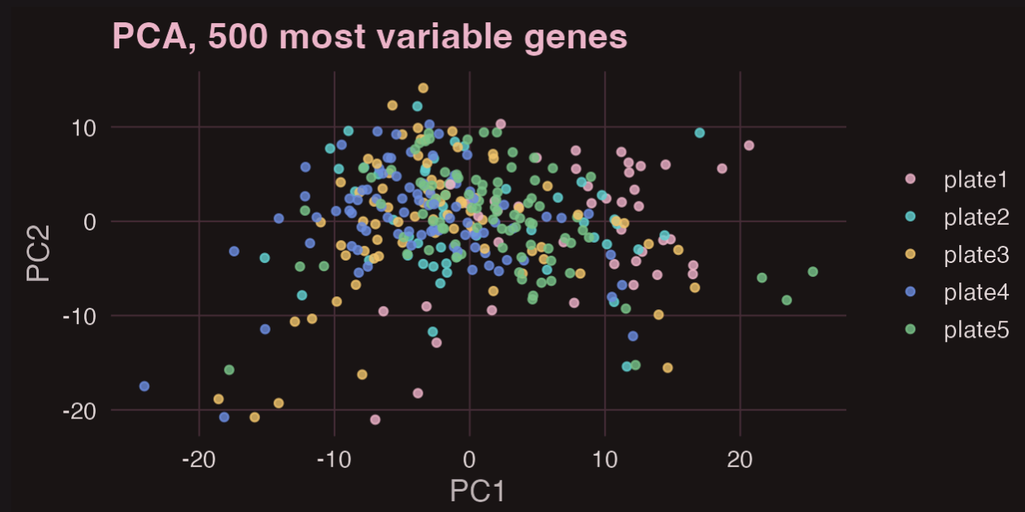
Hands-on · Data and Gene Selection

Exercises 1–3 · ≈ 10 min · hands-on notebook, sections 2–4

1. Colour the PCA by treatment, sex, and plate. Which one drives PC1 and PC2?
2. How much variance does `XIST` keep after adjustment? Compare correlations with 2,000 random genes.
3. How many lithium DEGs are there, and how many of them are among the 2,000 network genes?

Answer · Exercise 1 · What Drives the PCA?

```
raw_variance <- apply(expression_matrix, 1, var)
top500 <- names(sort(raw_variance,
                     decreasing = TRUE))[1:500]
pca <- prcomp(t(expression_matrix[top500, ]),
              scale. = TRUE)
# ANOVA p-value: does each variable explain PC1/PC2?
pc_p <- function(v) sapply(1:2, function(k)
  anova(lm(pca$x[, k] ~ sample_info[[v]]))[1, 5])
pc_pvals <- sapply(c("treatment", "sex", "plate"),
                   pc_p)
rownames(pc_pvals) <- c("PC1", "PC2")
signif(pc_pvals, 2)
##     treatment   sex   plate
## PC1     0.190 0.069 8.0e-16
## PC2     0.047 0.560 8.2e-01
```



Sequencing plate drives PC1 (p = 8e-16). Sex and treatment barely show on the first two PCs, even though `XIST` is the single most variable gene: one gene does not make a principal component. That is why plate and sex go into the model.

Answer · Exercise 2 · Adjust, Then Filter

```r
rank_of <- function(v, g)
  match(g, names(sort(v, decreasing = TRUE)))
v0 <- raw_variance; v1 <- gene_variance
xist <- rbind(
  variance = c(v0[["XIST"]], v1[["XIST"]]),
  rank = c(rank_of(v0, "XIST"),
           rank_of(v1, "XIST")))
colnames(xist) <- c("before", "after")
round(xist, 2)
##          before after
## variance  22.81  0.96
## rank       1.00 92.00
```

```r
set.seed(1)
random_genes <- sample(rownames(adjusted_matrix), 2000)
abs_cor <- function(g) {
  C <- cor(t(adjusted_matrix[g, ])); abs(C[upper.tri(C)])
}
pc1 <- function(g)
  prcomp(t(adjusted_matrix[g, ]), scale. = TRUE)$x[, 1]
neu <- sample_info$neutrophils
rbind(median_abs_r = c(median(abs_cor(selected_genes)),
                       median(abs_cor(random_genes))),
      pc1_vs_neutrophils = c(cor(pc1(selected_genes), neu),
                             cor(pc1(random_genes), neu))) |>
  `colnames<-`(c("top_2000", "random")) |> round(2)
##                   top_2000 random
## median_abs_r          0.15   0.20
## pc1_vs_neutrophils   -0.58   0.73
```

(a) Adjustment removes the female–male difference, so `XIST` loses most of its variance (22.8 → 0.96). It still varies between women, so it can stay among the top 2,000.

(b) The random genes correlate **more**, and the first PC of **both** sets follows the neutrophil fraction (a PC's sign is arbitrary: read |r|). In whole blood, the mix of cell types moves thousands of genes together. Strong correlation is not automatically interesting structure: ask what drives it.



Answer · Exercise 3 · How Much DE Reaches the Network?

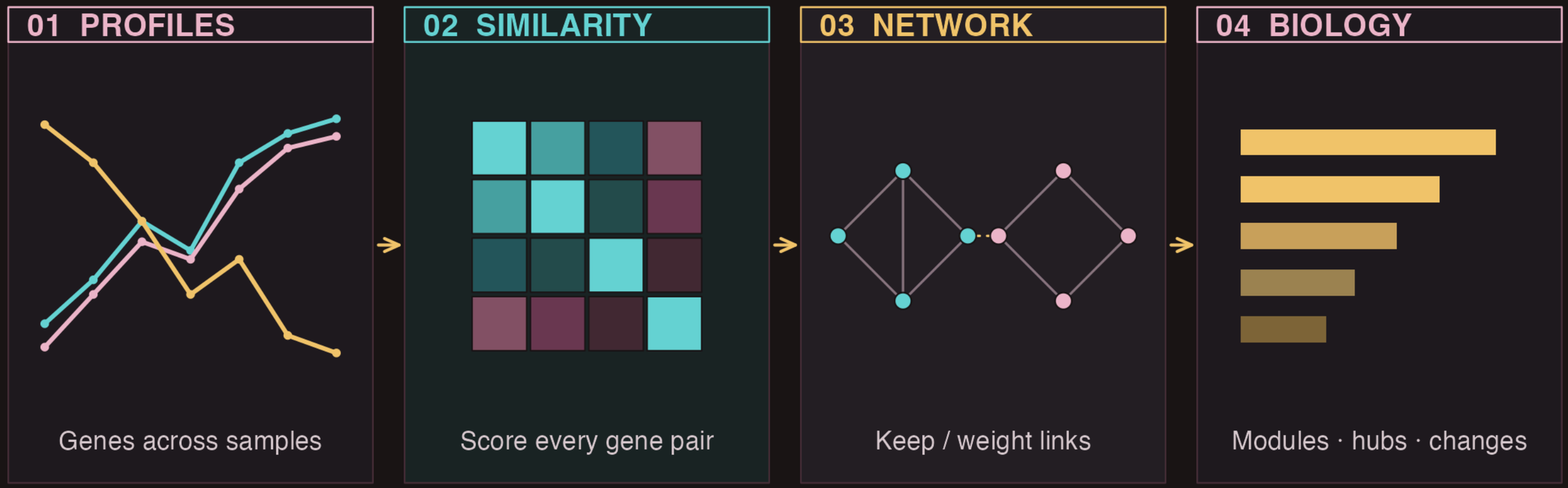
```
c(lithium_DEGs = sum(limma_results$adj.P.Val < 0.05),
  diagnosis_DEGs = sum(diagnosis_results$adj.P.Val < 0.05),
  in_network = sum(limma_results[selected_genes, "adj.P.Val"] < 0.05))
##   lithium_DEGs diagnosis_DEGs     in_network
##           4958            495            560
```

Thousands of genes pass FDR for lithium, mostly with small fold changes: with 338 samples, small shifts are detectable. The diagnosis contrast finds fewer. DEGs are **under-represented** among the most variable genes: 28% of the network genes vs 38% genome-wide. In blood, variance is dominated by differences between people (HLA genotype, cell mix), not by lithium.

Part III · Building Networks

Three ways to define a link · WGCNA · ARACNE · wTO

From Expression Data to Networks



The methods differ in **how they score a pair** (correlation, mutual information, shared neighbors) and **which links they keep**.

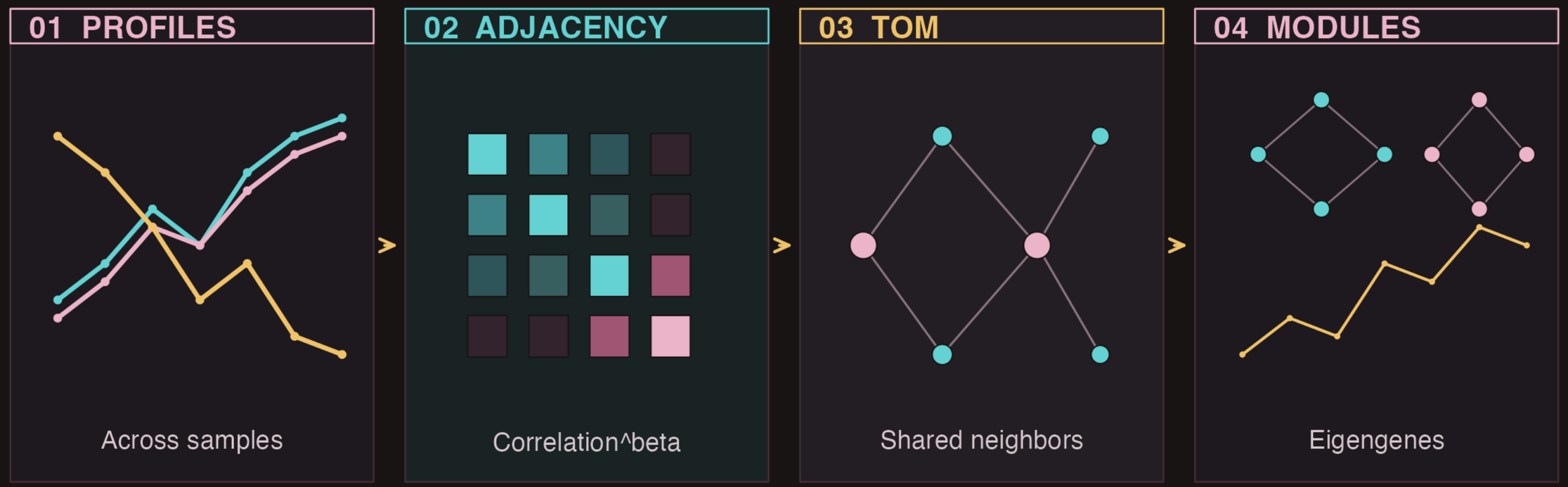
```
network_matrix <- adjusted_matrix[selected_genes, , drop = FALSE]  # plate/sex removed
dat_expr <- t(network_matrix)
dim(dat_expr)  # samples x genes for network methods
```

```
[1]  338 2000
```

WGCNA · From Correlation to Modules

WGCNA turns pairwise gene correlations into a weighted network, groups genes by shared neighbors, and summarizes each module with an eigengene.

$$a_{ij} = \left(\frac{1 + \mathrm{cor}(x_i, x_j)}{2} \right)^{\beta}$$



Output: modules and eigengenes for trait association. Boundaries depend on the gene set, soft power, and clustering settings; they are not functional annotations by themselves.

WGCNA · The Method, Step by Step

1 · Similarity. Signed: $s_{ij} = \frac{1+r_{ij}}{2}$ · unsigned: $s_{ij} = |r_{ij}|$

2 · Soft threshold. $a_{ij} = s_{ij}^{\beta}$: keep every link, but suppress weak ones. Choose β so that $\log P(k)$ vs $\log k$ is close to linear.

3 · Topological overlap.

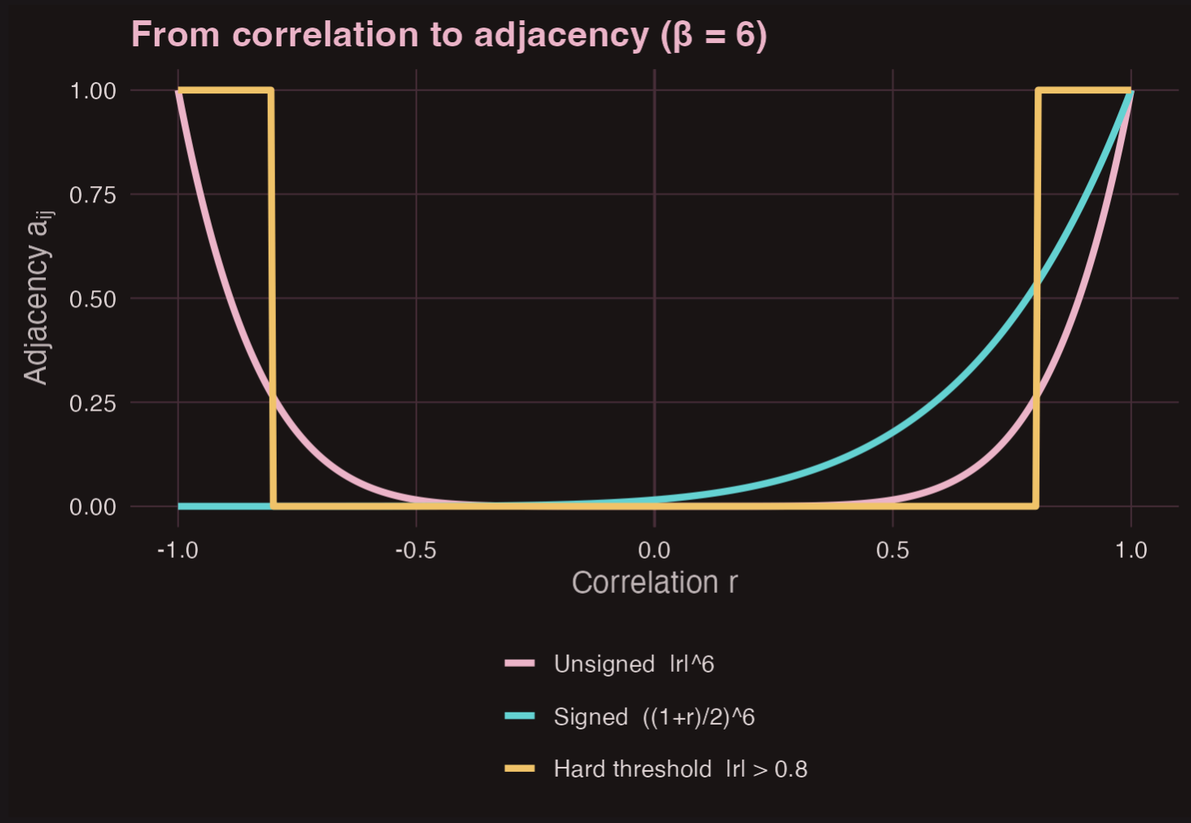
$$\mathrm{TOM}_{ij} = \frac{\sum_u a_{iu} a_{uj} + a_{ij}}{\min(k_i, k_j) + 1 - a_{ij}}$$

4 · Clustering. Average-linkage tree on $1 - \mathrm{TOM}$; **dynamic tree cut** defines modules (`minModuleSize`).

5 · Eigengene. ME_q = first principal component of the module's scaled expression. Modules with $\mathrm{cor}(\mathrm{ME}) > 1 -$ `mergeCutHeight` merge.

6 · Membership (kME). kME = *eigengene-based connectivity*: the correlation of gene i with its module's eigengene, $k\mathrm{ME}_i = \mathrm{cor}(x_i, \mathrm{ME}_q)$. Near 1, the gene follows the module closely; the highest are **intramodular hubs**.

WGCNA · Signed vs Unsigned Networks



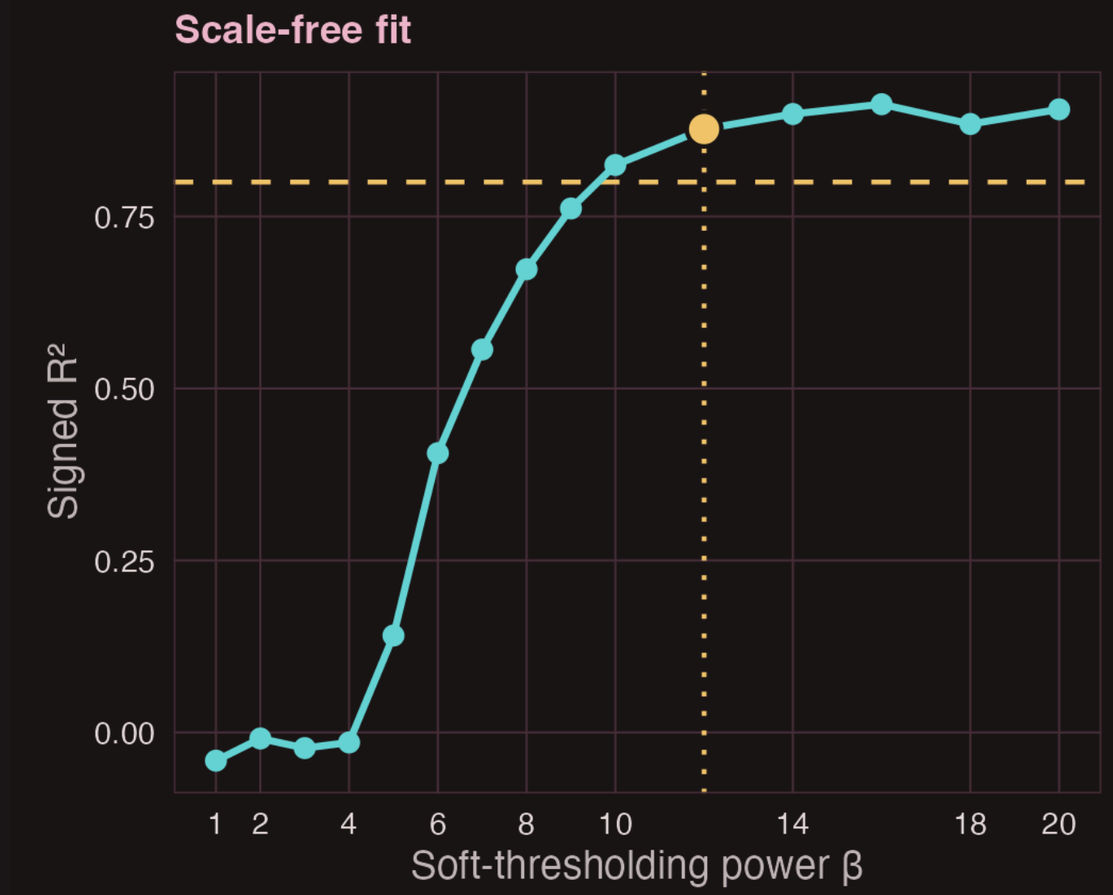
Unsigned: genes with $r = -0.9$ are as linked as those with $r = +0.9$. Modules mix opposite patterns.

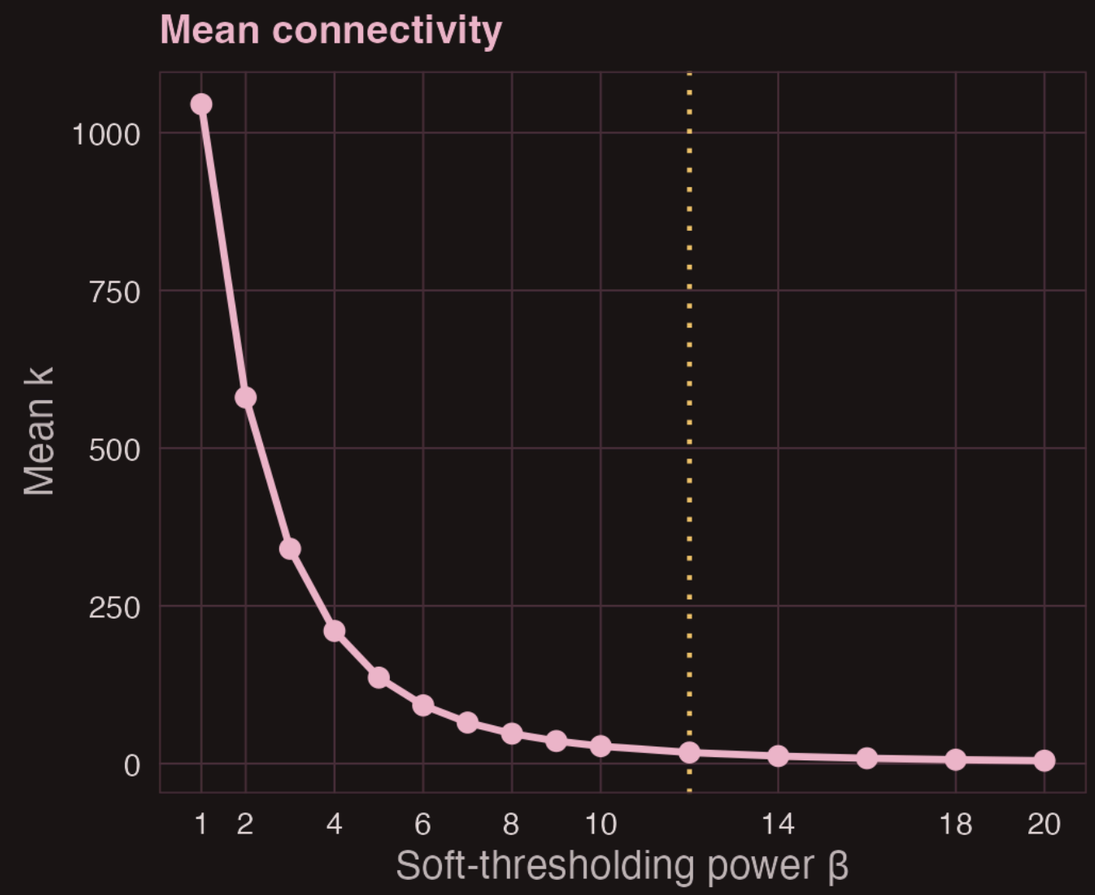
Signed: only positive co-expression produces strong links. Modules hold genes that **rise and fall together**, which makes eigengenes easier to interpret.

Soft vs hard: a hard cutoff throws away information and is sensitive to the cutoff; soft thresholding keeps a continuous weight.

We use **signed** networks.

WGCNA · Choosing a Soft-thresholding Power





Pick the lowest β where the fit plateaus near **$R^2 \approx 0.8$** without letting connectivity collapse. Here WGCNA's estimate is **$\beta = 12$** ($R^2 = 0.88$, mean k = 18), and that is what we use.

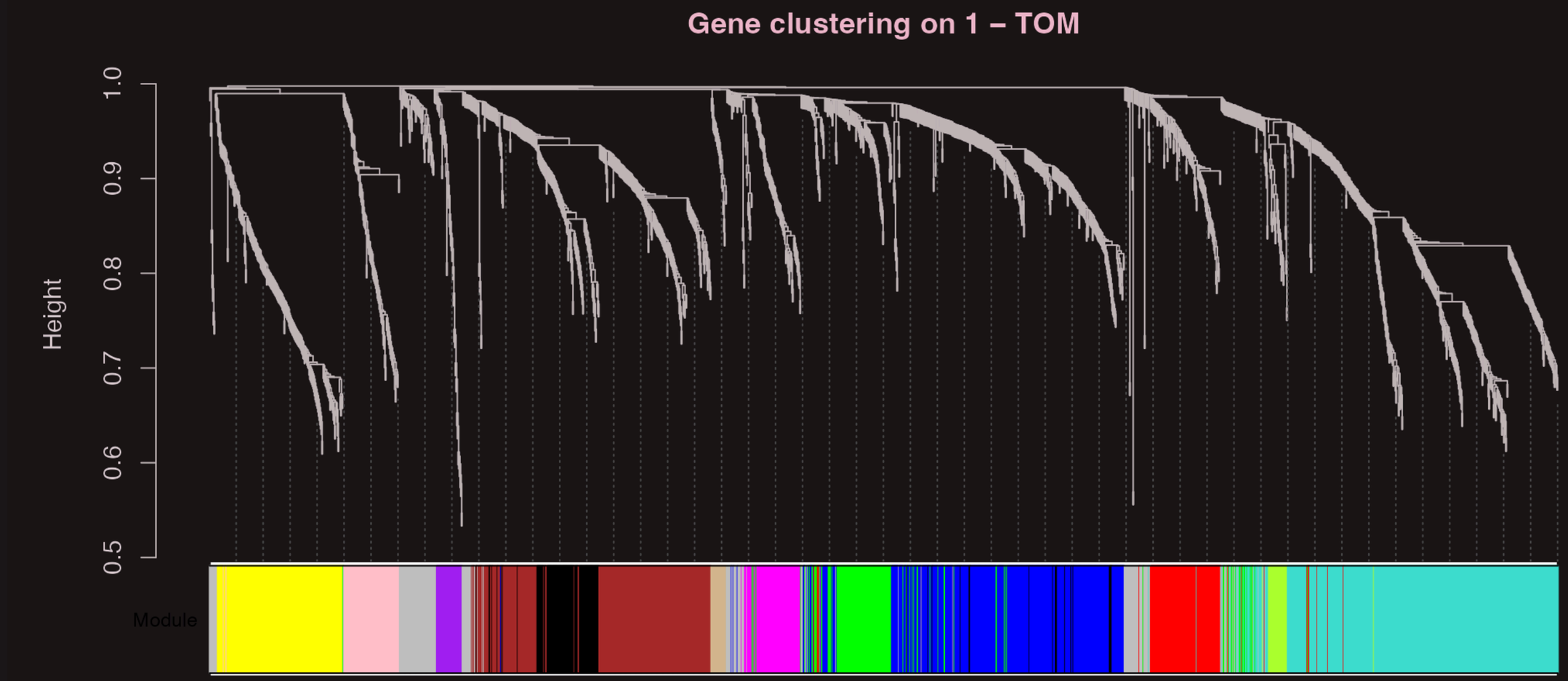
Run a Small, Reproducible WGCNA

```r
soft_power <- power_diagnostic$powerEstimate  # from the soft-threshold diagnostic
wgcna_fit <- WGCNA::blockwiseModules(
  dat_expr,
  power = soft_power,
  networkType = "signed",
  TOMType = "signed",
  minModuleSize = 20,
  mergeCutHeight = 0.25,
  randomSeed = 20261109,
  verbose = 0
)
module_colors <- as.character(wgcna_fit$colors)  # already colour names
names(module_colors) <- colnames(dat_expr)
module_eigengenes <- WGCNA::orderMEs(wgcna_fit$MEs)
table(module_colors)
```

```
module_colors
     black       blue      brown      green greenyellow       grey
       112        334        255        143         30        172
   magenta       pink     purple        red        tan  turquoise
        80         84         38        116         26        426
    yellow
       184
```

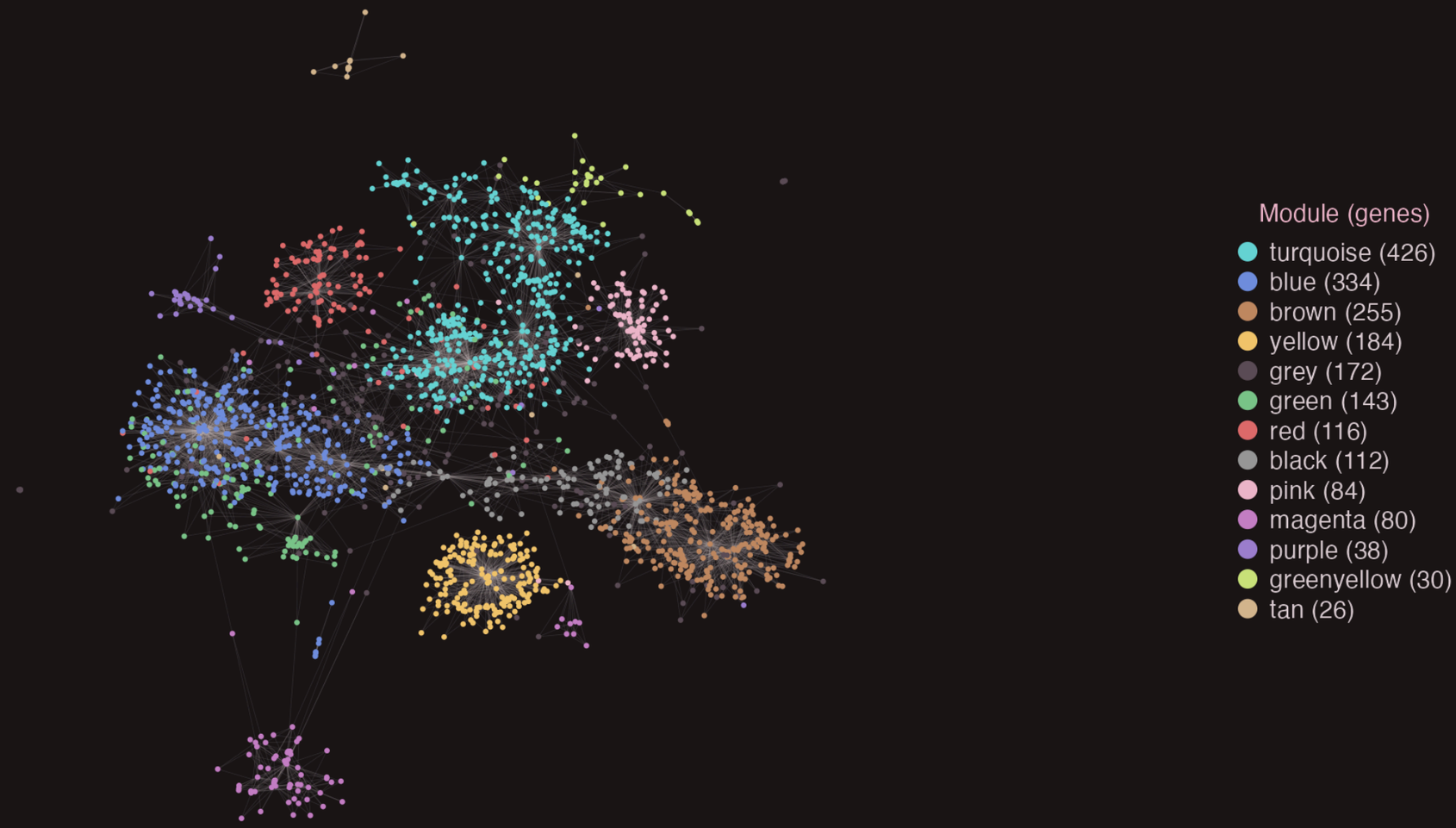
One network for everyone. WGCNA runs **once**, on all 338 samples (controls and patients together); the groups enter only afterwards, when module eigengenes are related to traits.

WGCNA · Visualize the Module Dendrogram



Each leaf is a gene; branches that merge low are genes with highly overlapping neighborhoods. The color strip is the module assignment from dynamic tree cut.

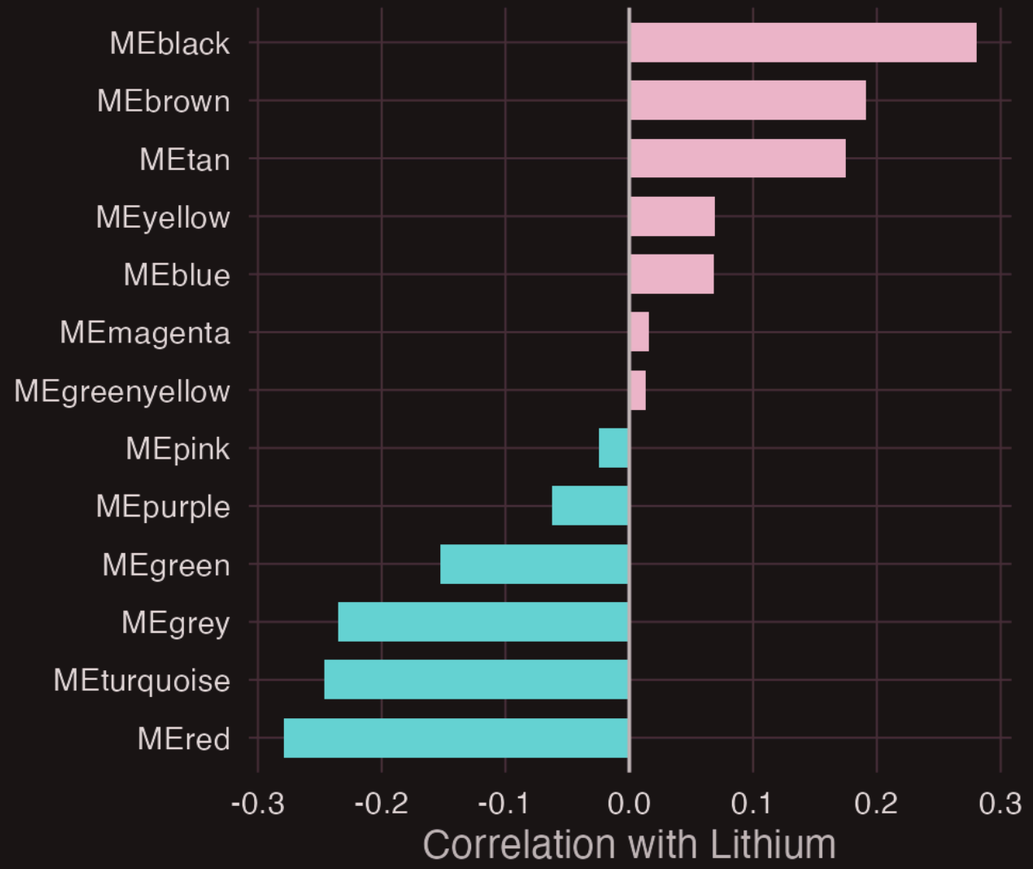
WGCNA · The Module Network



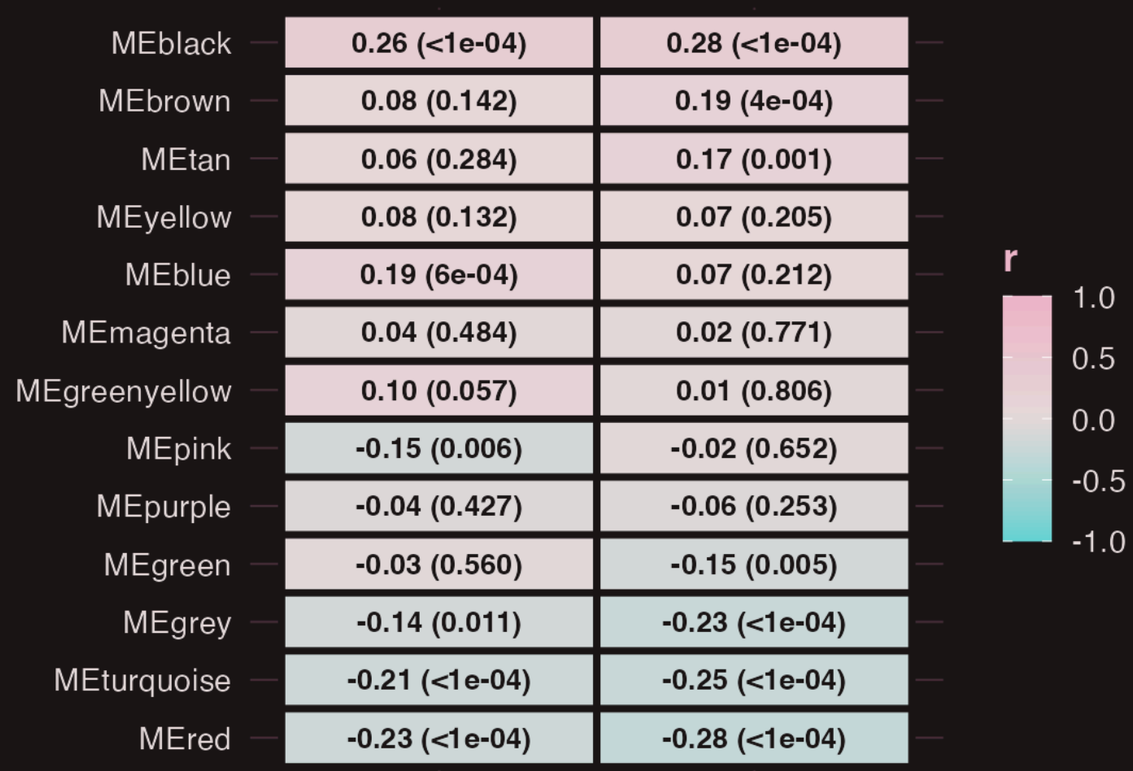
All 2,000 genes of the single network, each linked to its **3 strongest TOM partners**. Each colour is a module: modules appear as dense neighbourhoods, and the links between them are genes that share partners across modules.

WGCNA · Relate Modules to the Phenotype

Eigengene–trait correlation

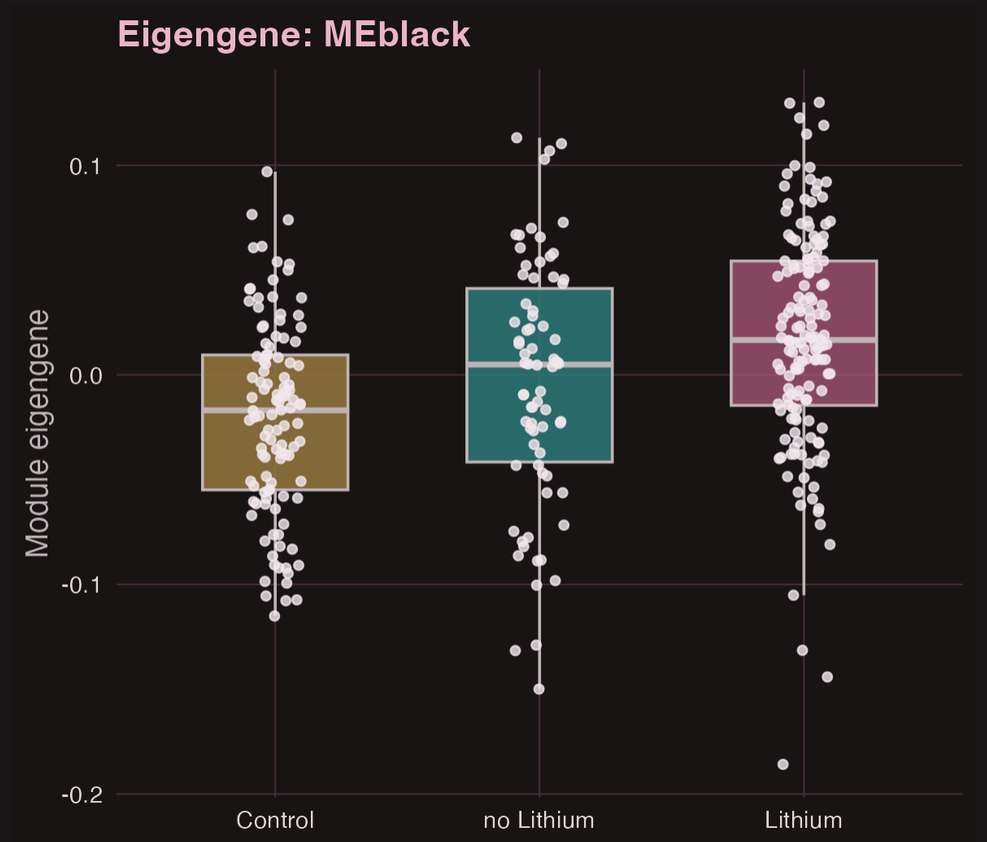


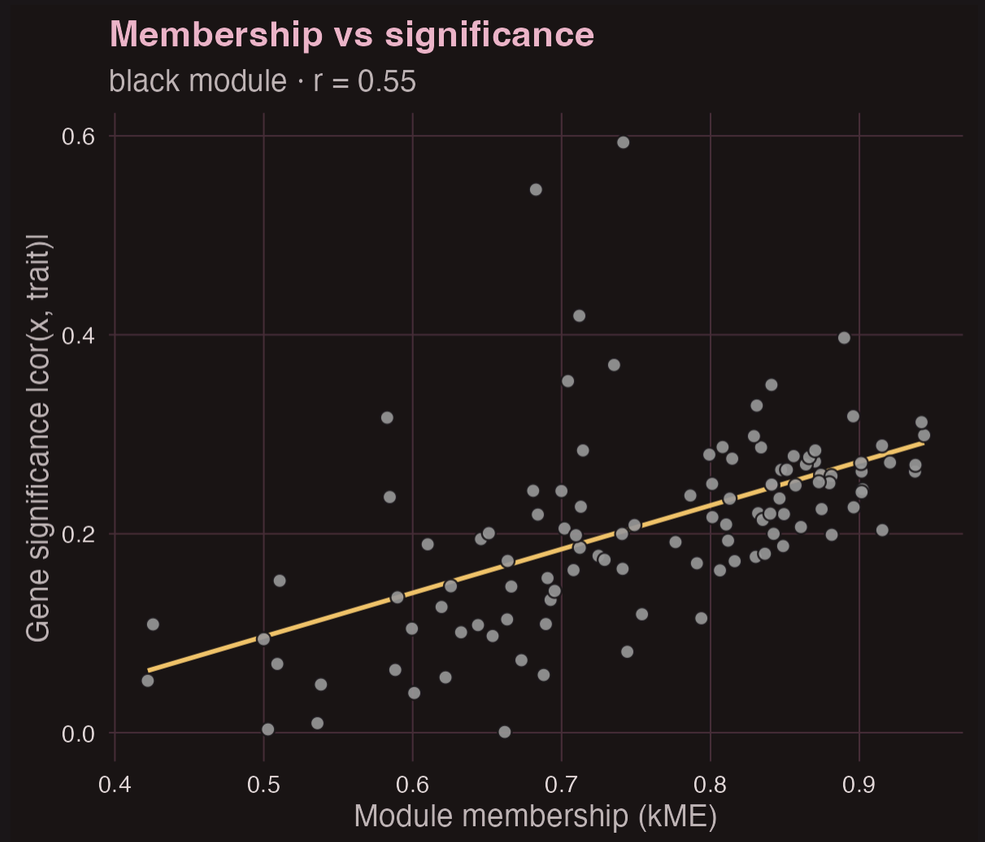
Module–trait heatmap



Two binary traits over all 338 samples: **bipolar disorder** (patients vs controls) and **lithium** (on lithium vs everyone else). Correlation with a 0/1 indicator is a two-group comparison of eigengenes: association, not causation.

WGCNA · Compare Eigengenes by Condition





Reading it: when kME and gene significance correlate positively, the module's most central genes are also the most trait-associated, which makes them candidate drivers. Here $r = 0.55$, so that pattern holds for the **black** module. These are the genes to follow up first. The grey bin holds unassigned genes and is never interpreted as a module.

WGCNA · Intramodular Hubs

Gene	kME	GS	logFC	FDR
RNF24	0.94	0.30	+0.25	0.0135
KDM6B	0.94	0.31	+0.29	0.0041
PXN	0.94	0.27	+0.19	0.0231
NOTCH1	0.94	0.26	+0.20	0.0213
FMNL1	0.92	0.27	+0.21	0.0255
GAB2	0.92	0.20	+0.14	0.0840
RASGRP4	0.92	0.29	+0.26	0.0030
KMT2D	0.90	0.24	+0.17	0.0831
C15orf39	0.90	0.26	+0.19	0.0421
NBEAL2	0.90	0.24	+0.14	0.1522

The ten genes most representative of the **black** module: highest **kME**, the correlation of each gene with the module eigengene. GS = gene significance, |cor(gene, lithium)|; logFC and FDR come from limma.

A hub summarizes how its module behaves; being central does not make it a regulator.

Hubs with **high kME but no DE signal** are exactly the genes a DEG-only analysis would miss.

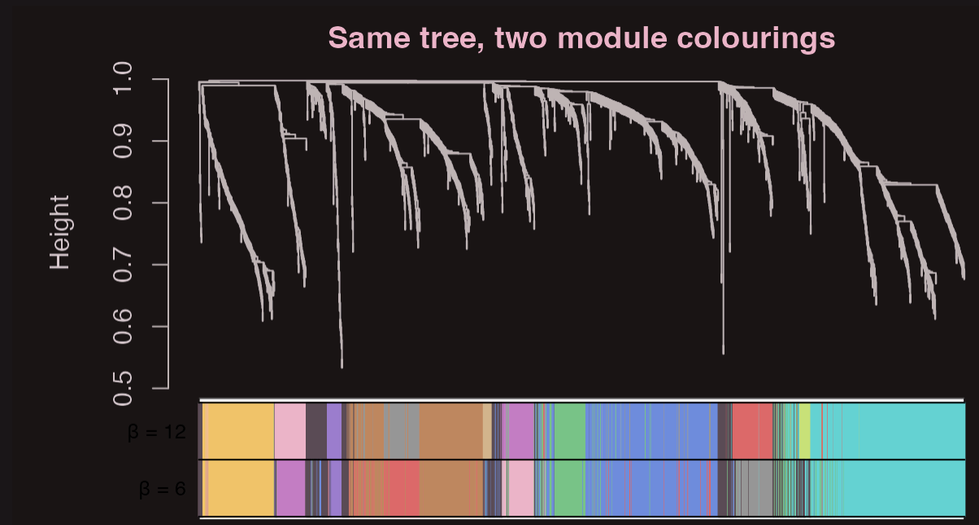
Hands-on · WGCNA

Exercises 4–5 · ≈ 10 min · hands-on notebook, section 5

4. Re-run WGCNA with `power = 6`. How many modules remain? How many genes become grey?
5. Which eigengene correlates most with lithium? List its 10 top-kME genes.

Answer · Exercise 4 · A Lower Soft Power

```
fit6 <- WGCNA::blockwiseModules(
  dat_expr, power = 6, networkType = "signed",
  TOMType = "signed", minModuleSize = 20,
  mergeCutHeight = 0.25, randomSeed = 20261109,
  verbose = 0)
fit6_colors <- as.character(fit6$colors)
n_mod <- function(m) length(setdiff(unique(m), "grey"))
data.frame(
  power = c(soft_power, 6),
  modules = c(n_mod(module_colors), n_mod(fit6_colors)),
  grey = c(sum(module_colors == "grey"),
           sum(fit6_colors == "grey")))
##   power modules grey
## 1    12      12  172
## 2     6      10  231
```



The power changes the partition: β = 6 gives **10** modules and **231** grey genes, against 12 and 172 at β = 12. Where one module ends and the next begins is partly a choice, so report the power you used.

Answer · Exercise 5 · The Lithium Module and Its Hubs

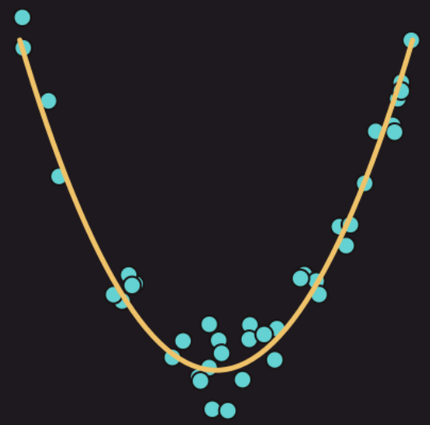
```
mt <- module_trait[module_trait$module != "MEgrey",
                   c("module", "correlation")]
head(mt[order(-abs(mt$correlation)), ], 3)
##        module correlation
## 4     MEblack   0.2807353
## 6       MEred  -0.2792545
## 2 MEturquoise  -0.2460128
kme_top <- setNames(kme_matrix[[paste0("kME", top_color)]],
                    rownames(kme_matrix))
hubs <- head(sort(kme_top[module_colors == top_color],
                  decreasing = TRUE), 10)
round(hubs, 2)
##    RNF24    KDM6B      PXN   NOTCH1    FMNL1     GAB2
##     0.94     0.94     0.94     0.94     0.92     0.92
##  RASGRP4    KMT2D C15orf39   NBEAL2
##     0.92     0.90     0.90     0.90
```

The **black** eigengene correlates most with lithium (r = 0.28). Its top-kME genes, led by **RNF24**, best represent the module's expression pattern.

They are the first genes to follow up, but being central does not make a gene a regulator.

ARACNE · How It Works

01 DEPENDENCE



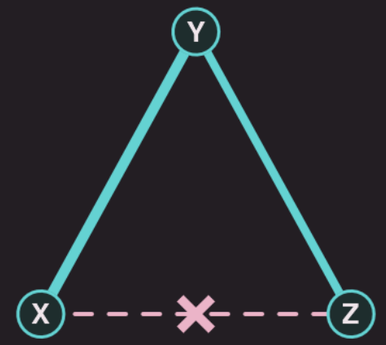
Linear and nonlinear

02 MUTUAL INFO



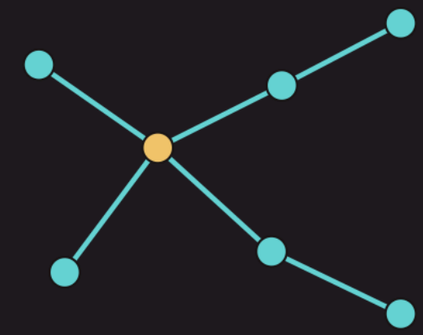
I(X;Y) for every pair

03 DPI



Drop weakest of triplet

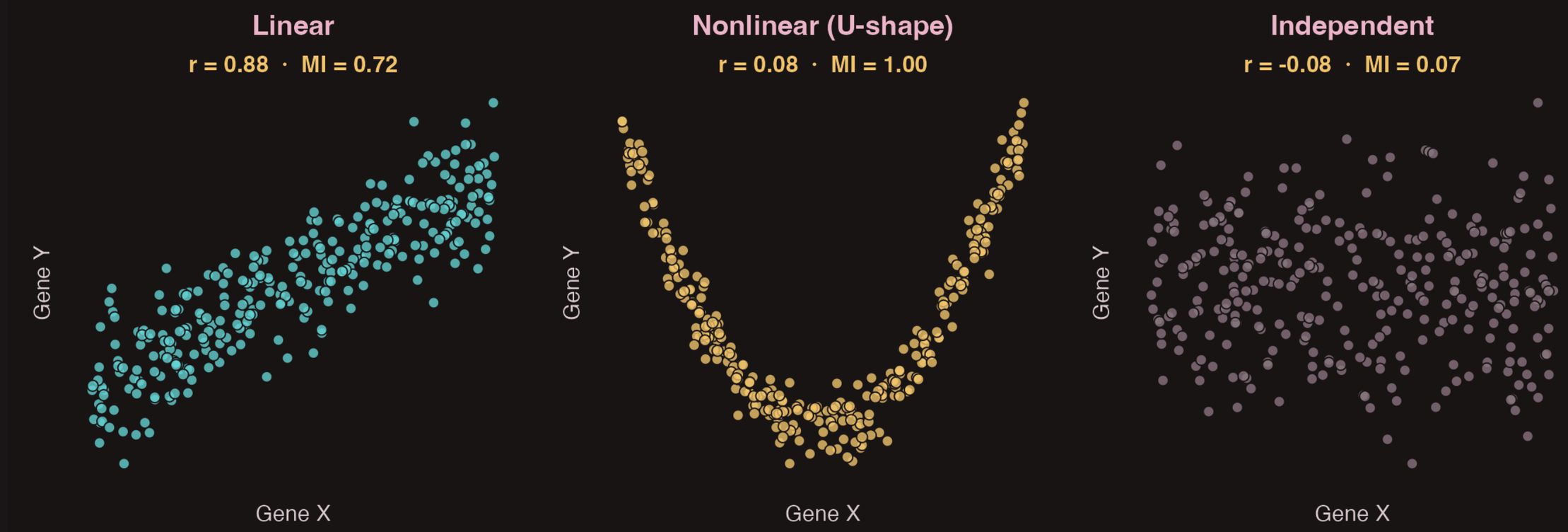
04 SPARSE NET



Direct candidates

What it asks: which gene pairs show statistical dependence, including nonlinear dependence? Mutual information (MI) scores every pair, and the data processing inequality (DPI) prunes likely indirect links. The result is a sparse, undirected network. It does not show causal direction.

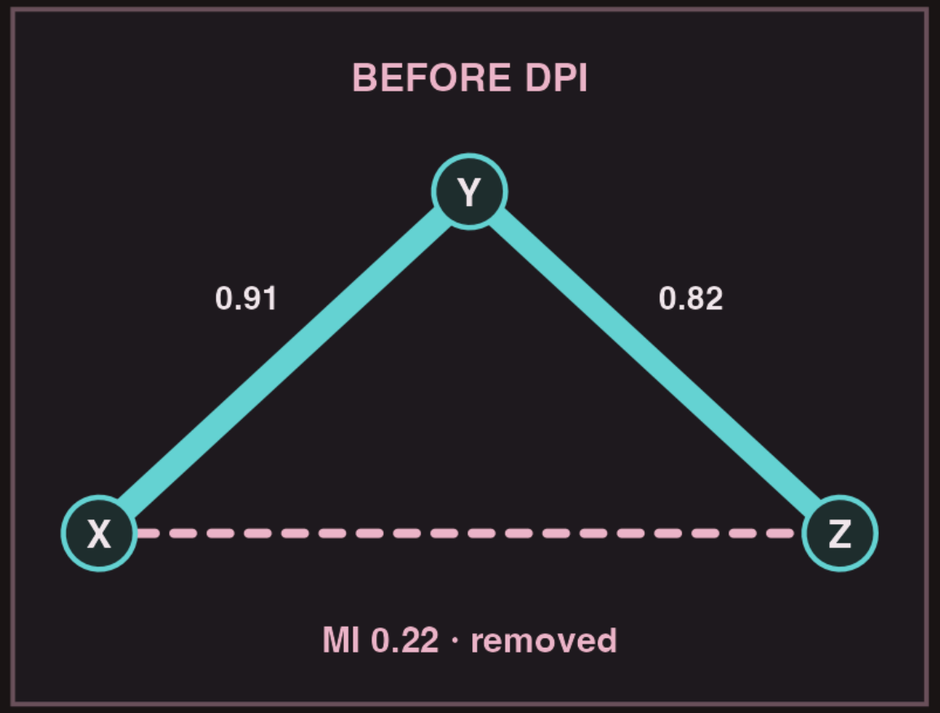
ARACNE · Correlation Can Miss Dependence

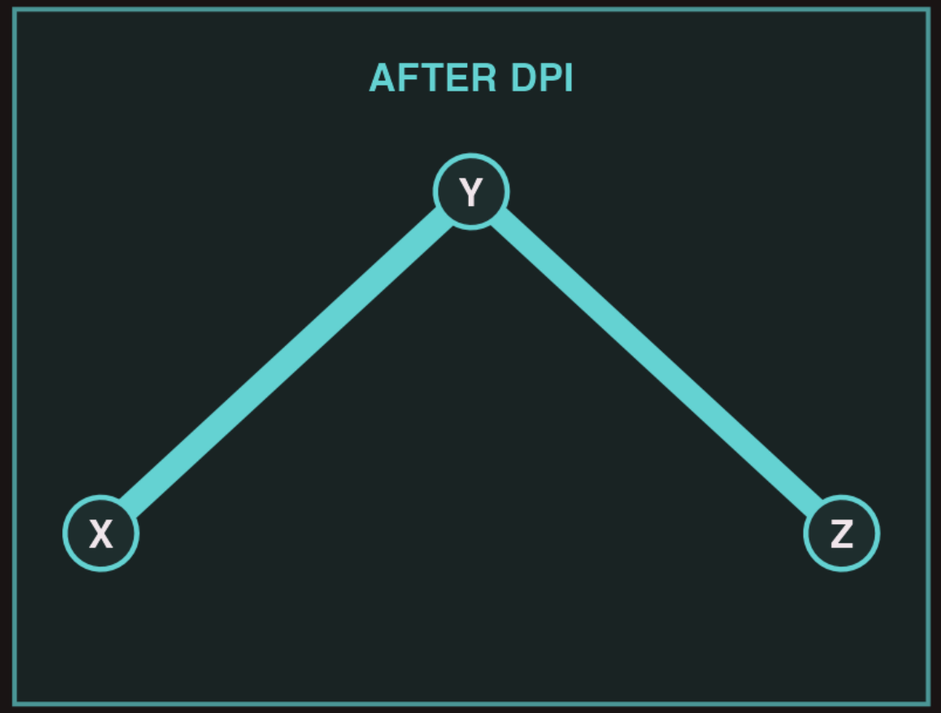


$$I(X;Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)\,p(y)} \qquad I = 0 \iff X \perp Y$$

MI is zero only under independence, so it detects the U-shape that correlation misses. The cost is that MI needs **more samples** to estimate reliably.

ARACNE · The Data Processing Inequality





If $X \to Y \to Z$ is a chain, information about Z in X cannot exceed what flows through Y:

$$I(X;Z) \le \min\{I(X;Y),\ I(Y;Z)\}$$

ARACNE inspects **every triplet** and removes its weakest edge, unless it is within a tolerance `eps` of the others. With `estimator = "spearman"`, minet uses the Gaussian form $I = -\frac{1}{2}\log(1-\rho^2)$.

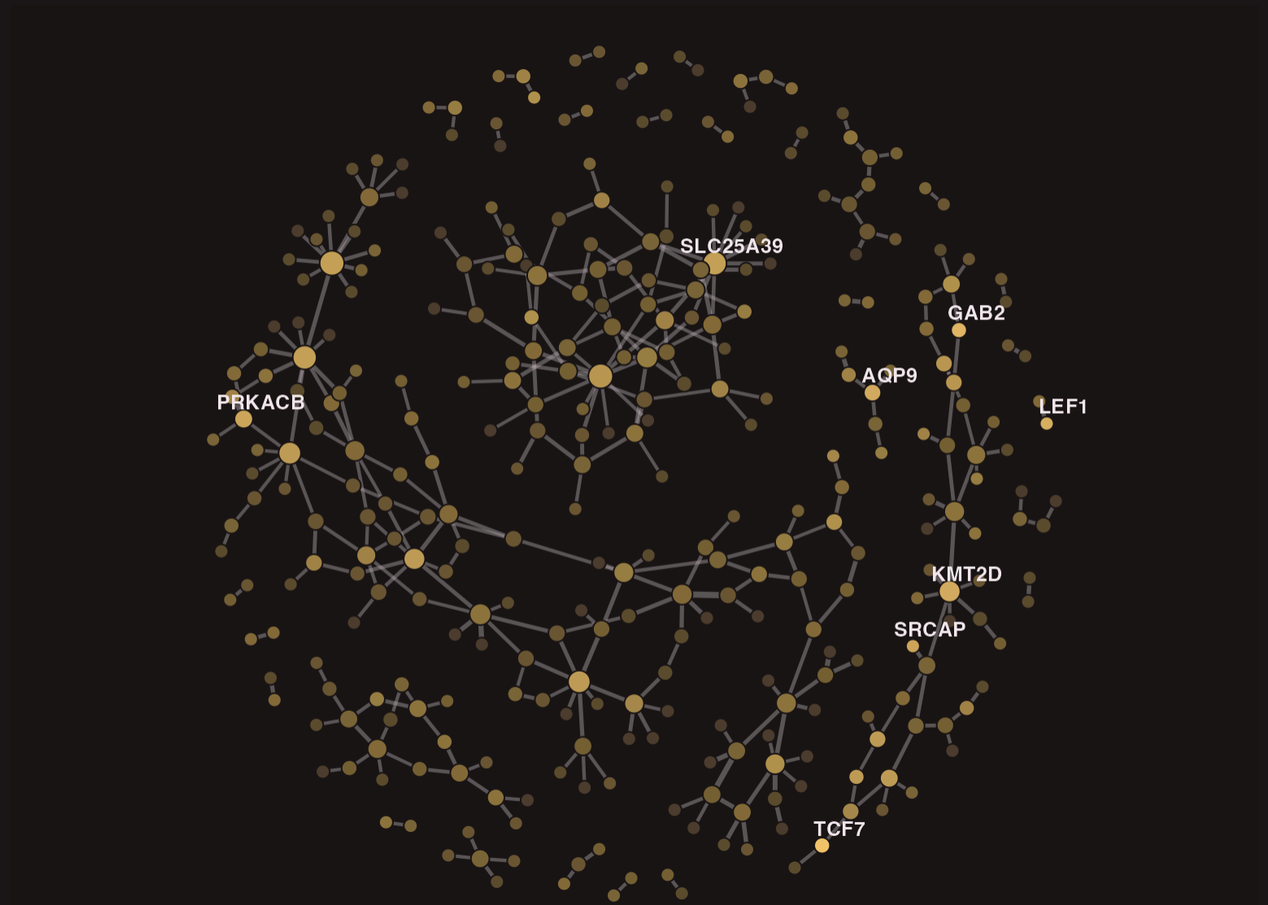
ARACNE · Run Mutual-Information Inference

```r
mi_matrix <- minet::build.mim(dat_expr, estimator = "spearman")  # samples x genes
aracne_matrix <- minet::aracne(mi_matrix, eps = 0)
c(edges = sum(aracne_matrix > 0) / 2,
  density = round(mean(aracne_matrix[upper.tri(aracne_matrix)] > 0), 4))
```

```
   edges   density
2685.0000    0.0013
```

One network from all 338 samples, as with WGCNA. **Input:** samples x genes. `build.mim()` estimates pairwise MI; `aracne()` applies DPI. **Output:** a sparse, symmetric adjacency matrix of MI values. Read each edge as statistical dependence, not regulation.

ARACNE · The Inferred Network



The strongest **350 ARACNE edges** of the single network.

Node **colour** = degree in the full ARACNE network; size = degree among the edges shown. Labels mark the 8 biggest **hubs**.

DPI keeps only **direct** dependencies, so the network is sparse and tree-like: hubs are genes with many non-redundant partners.

Hands-on · ARACNE

Exercise 6 · ≈ 6 min · hands-on notebook, section 6

6. Take the 50 genes with the highest ARACNE degree. Which WGCNA modules are they in? How many are also top-50 kME genes of the trait-associated module?

Answer · Exercise 6 · ARACNE Hubs vs WGCNA Hubs

```r
aracne_hubs <- names(sort(rowSums(aracne_matrix > 0),
                          decreasing = TRUE))[1:50]
sort(table(module_colors[aracne_hubs]), decreasing = TRUE)
##
##      blue turquoise     brown     black      grey
##        12        11         6         4         4
##    yellow     green   magenta       red       tan
##         4         3         3         2         1
kme_top50 <- names(head(sort(kme_top[module_colors == top_color],
                             decreasing = TRUE), 50))
length(intersect(aracne_hubs, kme_top50))
## [1] 4
```

4 genes are hubs in both. The ARACNE hubs spread over **10** WGCNA modules.

The methods define importance differently: a WGCNA hub sits at the centre of one module; DPI keeps only **direct** links, so ARACNE hubs have many non-redundant partners, often bridging modules. Genes found by both are the more robust candidates.

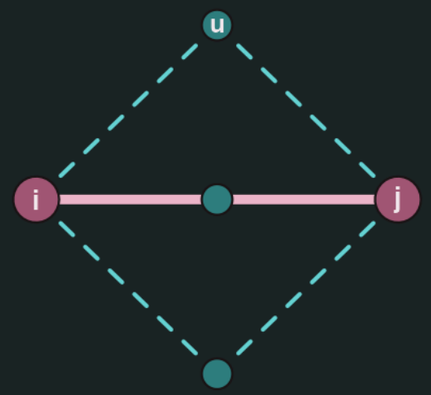
wTO · How It Works

01 SIGNED CORR.



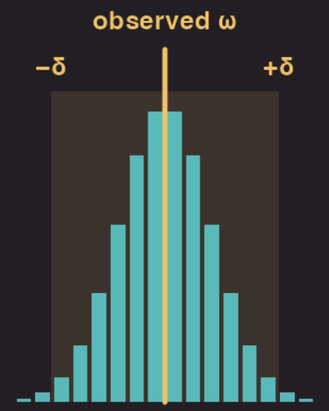
$a_{ij} \in [-1, 1]$

02 SHARED NBRS



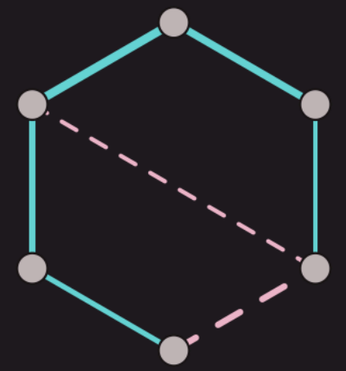
$\sum_u a_{iu} a_{uj}$

03 BOOTSTRAP



Resample samples, recompute ω

04 ROBUST EDGES



Signed wTO + p-value

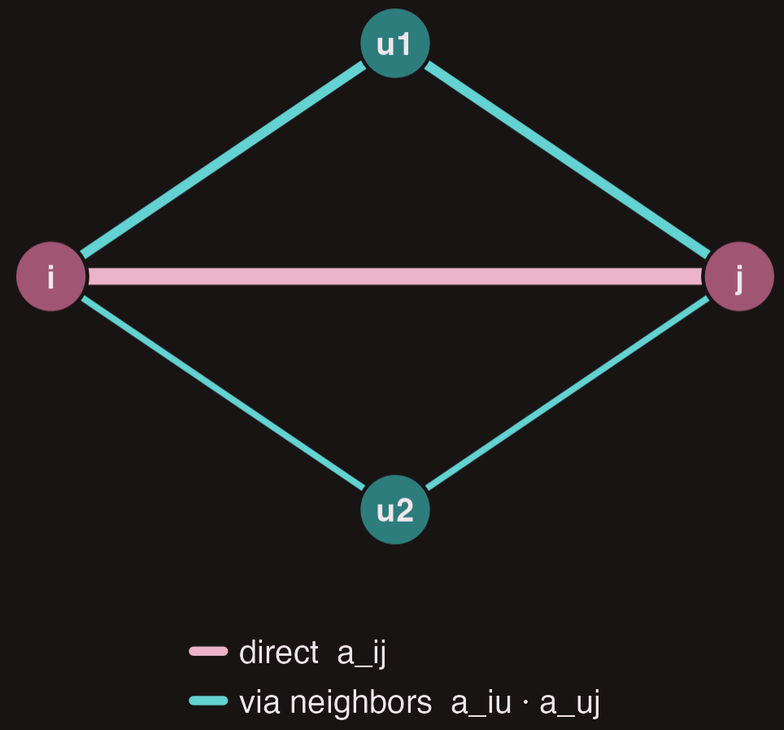
Idea: two genes are strongly linked when they correlate **and** share many neighbors with the same sign. Unlike TOM, wTO keeps the **sign**, so positive and negative co-regulation stay distinguishable.

wTO · The Formula

$$\omega_{ij} = \frac{\sum_{u \neq i,j} a_{iu}\, a_{uj} + a_{ij}}{\min(k_i, k_j) + 1 - |a_{ij}|}$$

with $a_{ij} \in [-1, 1]$ the signed correlation and $k_i = \sum_u |a_{iu}|$.

- $\omega_{ij} \in [-1, 1]$; the sign tells **concordant** vs **opposite** co-regulation.
- Neighbors **reinforce** the link when $a_{iu}a_{uj}$ agrees in sign with a_{ij}; they **weaken** it otherwise.
- **Bootstrap:** resample the samples with replacement and recompute ω many times. The p-value is the share of resamples in which ω moved by more than δ: small = a **stable** link.



wTO · Estimate Signed Topological Overlap

```r
source("R/wto_fast.R")  # vectorised wTO::wTO.fast(); identical output, much faster
# One network per group, all 2,000 genes, 100 bootstrap resamples.
# Built once (~15 min) and cached in results/wto_cache_2k.rds.
wto_nets <- load_wto_networks(network_matrix, sample_info, n = 100)
wto_control <- wto_nets$control
wto_nolithium <- wto_nets$no_lithium
wto_lithium <- wto_nets$lithium
sapply(list(control = wto_control, no_lithium = wto_nolithium, lithium = wto_lithium),
       function(x) c(pairs = nrow(x), stable = sum(x$pval.adj < 0.05)))
```

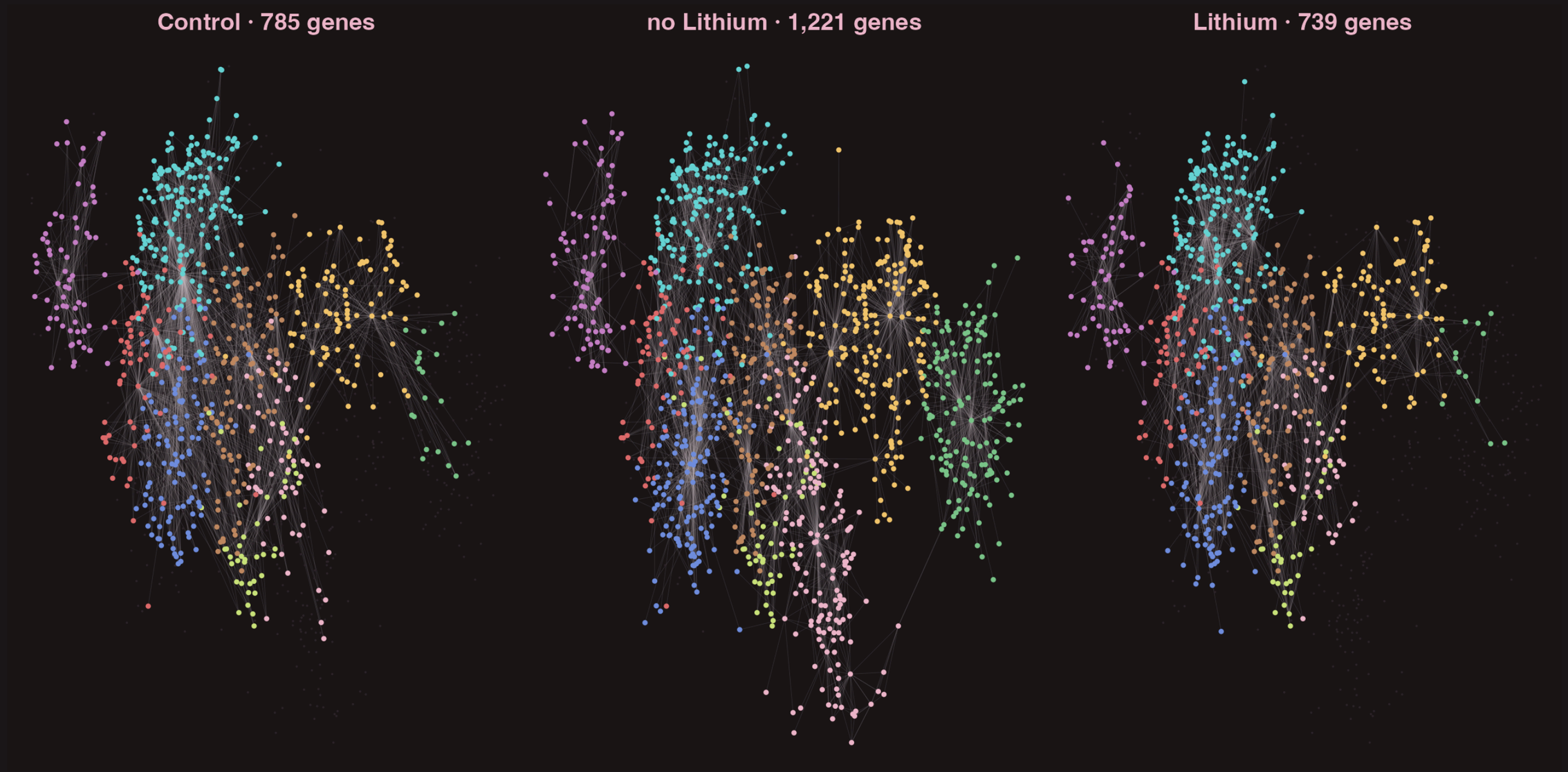
```
       control no_lithium lithium
pairs  1999000    1999000 1999000
stable 1868928    1244632 1955501
```

```r
head(wto_lithium, 3)
```

```
    Node.1   Node.2   wTO  pval pval.adj
    <char>   <char> <num> <num>    <num>
1: HLA-DQB1 HLA-DQA1 0.123     0        0
2: HLA-DQB1 HLA-DRB1 0.077     0        0
3: HLA-DQA1 HLA-DRB1 0.113     0        0
```

`wto_fast()` reproduces `wTO::wTO.fast()` exactly (same bootstrap, same p-values); it only vectorises one internal step. The p-value is **stability**: the share of resamples in which ω moved by more than $\delta = 0.2$.

wTO · Modules in Each Group



Only links with bootstrap $p_{adj} < 0.001$ **and** $|\omega| > 0.4$; each gene is drawn with its **3 strongest partners**, as in the WGCNA module network. Colours = 9 **communities** (Louvain) found on the three groups together, so the same colour means the same module in every panel. Watch which modules **appear, shrink, or vanish** between groups.

How to Read a wTO Edge

Field	What it records
`Node.1`, `Node.2`	The gene pair
`wTO`	Signed weighted topological overlap, $\omega_{ij} \in [-1, 1]$
`pval`	Share of bootstrap resamples in which ω_{ij} moved by more than δ
`pval.adj`	BH-adjusted `pval`: small means a **stable** link

`wTO.Complete()` adds signed/absolute variants and resampling choices; `wTO.fast()` returns these five columns. Stability and strength are different: inspect both before setting an edge threshold.

Hands-on · wTO

Exercise 7 · ≈ 8 min · hands-on notebook, section 7

7. List the 10 strongest stable links in each condition. How many pairs are stable in both but **flip sign**?

Answer · Exercise 7 · Strongest Links and Sign Flips

```r
stable <- function(x) filter(x, pval.adj < 0.05)
top_links <- function(x) stable(x) |>
  arrange(desc(abs(wTO))) |> head(3) |>
  transmute(link = paste(Node.1, Node.2, sep = "–"), wTO)
top_links(wto_control)
##            link   wTO
##          <char> <num>
## 1:  RPL11–ZNF22 0.499
## 2: ZNF22–NDUFA5 0.497
## 3: ZNF22–ZNF761 0.497
top_links(wto_lithium)
##             link   wTO
##           <char> <num>
## 1:    ZNF22–SNX4 0.477
## 2: ZCCHC10–ZNF22 0.473
## 3:  GIMAP2–ZNF22 0.473
both <- inner_join(stable(wto_control), stable(wto_lithium),
                   by = c("Node.1", "Node.2"),
                   suffix = c(".ctrl", ".li"))
c(stable_in_both = nrow(both),
  sign_flips = sum(sign(both$wTO.ctrl) != sign(both$wTO.li)))
## stable_in_both     sign_flips
##        1854065         279219
```

279,219 of the 1,854,065 pairs that are stable in both controls and lithium-treated patients **flip sign**.

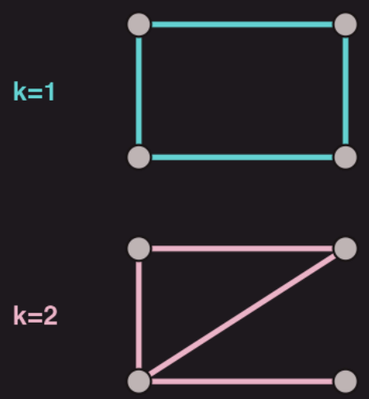
Sign flips are the raw material of CoDiNA's **β** class. Lithium was not randomized, so treat them as hypotheses.

Part IV · Comparing Networks

Consensus: what is robust? · CoDiNA: what changed?

Consensus Networks · How It Works

01 COHORTS



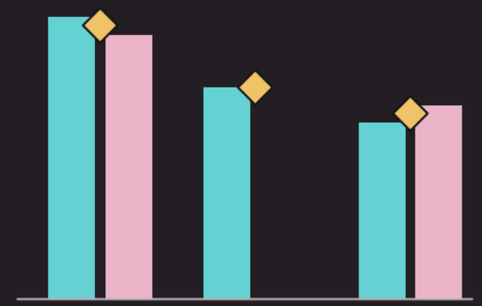
Related cohorts, k = 1…n

02 ALIGN PAIRS



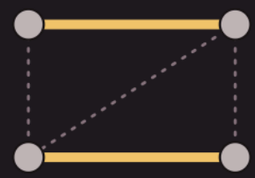
Common genes & pairs

03 WEIGHT



$\alpha^k \propto |\omega^k|$

04 CONSENSUS



Conserved core

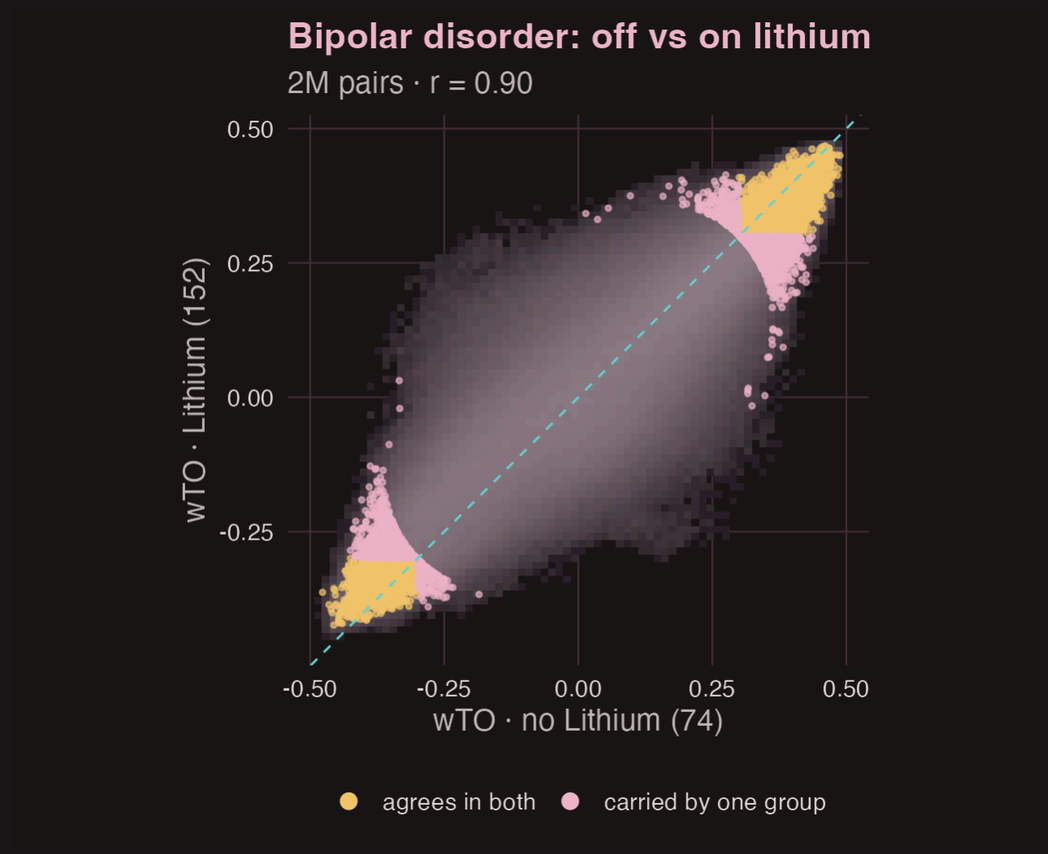
$$\Omega_{ij} = \sum_{k=1}^{n} \alpha_{ij}^{k}\, \omega_{ij}^{k}, \qquad \alpha_{ij}^{k} = \frac{|\omega_{ij}^{k}|}{\sum_{k} |\omega_{ij}^{k}|}$$

Each dataset contributes in proportion to its edge strength, so a link that is strong everywhere stays strong. `wTO.Consensus()` also combines the per-cohort p-values with **Fisher's method**.

Consensus · A Worked Example

```r
as_wto_input <- function(x) data.frame(
  Node.1 = factor(x$Node.1), Node.2 = factor(x$Node.2),
  wTO = x$wTO, pval = x$pval)
# bipolar network: patients off + on lithium
consensus <- wTO::wTO.Consensus(list(
  as_wto_input(wto_nolithium),
  as_wto_input(wto_lithium)))
# |wTO| peaks near 0.4 here
robust <- consensus |>
  mutate(padj = p.adjust(pval.fisher, "BH")) |>
  filter(abs(CN) >= 0.3, padj < 0.05)
nrow(robust)
```

```
[1] 275044
```



We combine the networks of patients **off** (74) and **on** (152) lithium into one **bipolar-disorder consensus**: links that hold whatever the treatment. Of the 275,044 consensus links, **188,173 agree in both** and **86,871 are carried by one group**: weighting by $|\omega|$ lets the stronger network dominate. **Check per-network agreement.**

Hands-on · Consensus

Exercise 8 · ≈ 6 min · hands-on notebook, section 8

8. Build the bipolar consensus (off + on lithium). How many links have |CNI| ≥ 0.3 and BH-adjusted Fisher p < 0.05? How many agree in both groups?

Answer · Exercise 8 · The Bipolar Consensus

```
nrow(robust)   # |CN| >= 0.3 and BH-adjusted Fisher p < 0.05
## [1] 275044
keys <- mutate(robust, across(c(Node.1, Node.2), as.character))
ab_agree <- inner_join(wto_nolithium, wto_lithium,
                       by = c("Node.1", "Node.2"),
                       suffix = c(".off", ".on")) |>
  semi_join(keys, by = c("Node.1", "Node.2"))
with(ab_agree, table(agrees = sign(wTO.off) == sign(wTO.on) &
                       pmin(abs(wTO.off), abs(wTO.on)) >= 0.3))
## agrees
##  FALSE   TRUE
##  86871 188173
```

275,044 consensus links pass; only part of them are strong and same-signed in **both** patient groups.

Weighting by |ω| lets the stronger network dominate: a link strong in one group and weak in the other still gets a large CN and a small Fisher p. The links that agree form the shared bipolar network; the ones carried by one group are where CoDiNA looks for treatment effects.

Consensus vs Differential: Two Different Questions

Consensus

*"What is **robust** within the same condition across independent datasets?"*

- Input: networks of the **same** biology (cohorts, platforms, batches)
- Output: conserved edges with combined weight and p-value
- Use to build a reliable reference network

Differential (CoDiNA)

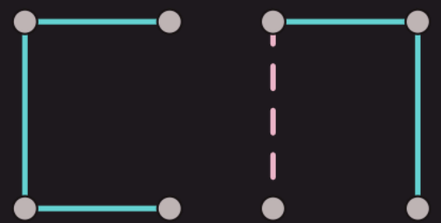
*"What **changed** between conditions?"*

- Input: networks of **different** conditions on the same genes
- Output: links classified as conserved, inverted, or specific
- Use to find candidate rewiring

> A robust analysis often uses both: build a **consensus** network within each condition, then compare conditions with **CoDiNA**.

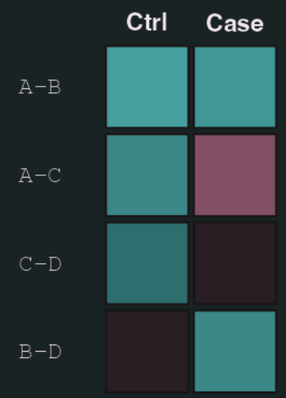
CoDiNA · How It Works

01 NETWORKS



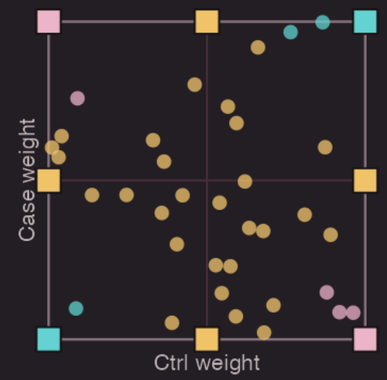
One per condition

02 UNION



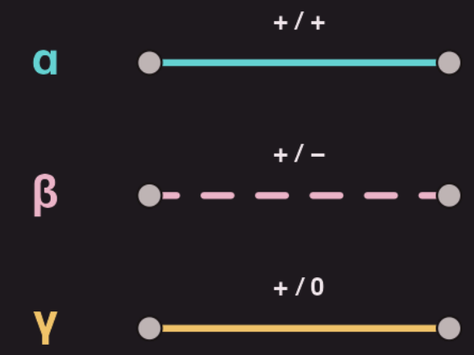
Same pairs, missing = 0

03 GEOMETRY



Distance to ideal vertices

04 CLASSES



α · β · γ + scores

Compare the same gene pairs: α = present in all networks with the same sign; **β** = present in all networks with opposite signs; **γ** = present in some networks only. Here **three** networks: controls (112), patients off lithium (74), and on lithium (152). Lithium was not randomized: treat changes as candidates, not proof of rewiring.

CoDiNA · Edge Classes



For two networks, **β edges** are often the most interesting: the same pair is co-expressed in both conditions, but the relationship **flips sign**.

CoDiNA · The Method, Step by Step

1 · Union. Put all networks on the same set of pairs; a missing link gets weight 0. Optionally rescale with `stretch`.

2 · Discretize. Map each weight to $\{-1, 0, +1\}$ using `cutoff`, so every link becomes a vertex of the cube $[-1, 1]^n$.

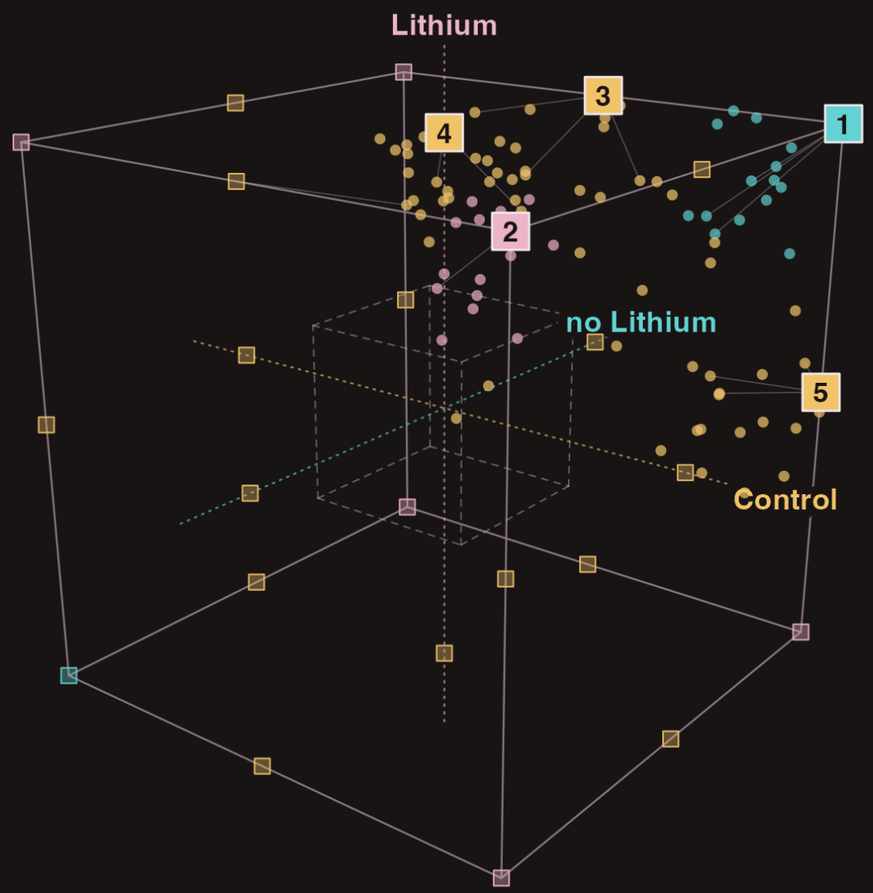
3 · Classify. Φ = α, β, or γ; $\tilde{\Phi}$ adds *where*, e.g. `g.Lithium` = γ present only in the lithium network.

4 · Score. Each link is a point $\mathbf{w}$ in $[-1, 1]^n$:

- `Score_center` = $\|\mathbf{w}\|$: strength, distance from the origin
- `Score_internal` = distance to its ideal vertex
- `Score_ratio` = strength within $\tilde{\Phi}$ ÷ `Score_internal`

5 · Prioritize. High `Score_ratio` means a **strong, well-classified** link. `ClusterNodes()` then assigns genes to classes from their links.

CoDiNA · Three Networks, One Cube



Each link has one weight per network, so with **three** networks it is a point $(w_{\text{Ctrl}}, w_{\text{off}}, w_{\text{on}})$ in the cube $[-1, 1]^3$. Its class is the **nearest ideal corner**:

	Corner	Class	Reading
1	(+, +, +)	`a`	conserved
2	(+, −, +)	`b.noLithium`	sign flips off lithium
3	(0, +, +)	`g.noLithium.Lithium`	**disease**
4	(0, 0, +)	`g.Lithium`	**lithium only**
5	(+, +, 0)	`g.Control.noLithium`	**lost on lithium**

26 corners: **2 α**, **6 β**, **18 γ**. The dashed inner box is the cutoff: weights inside count as 0.

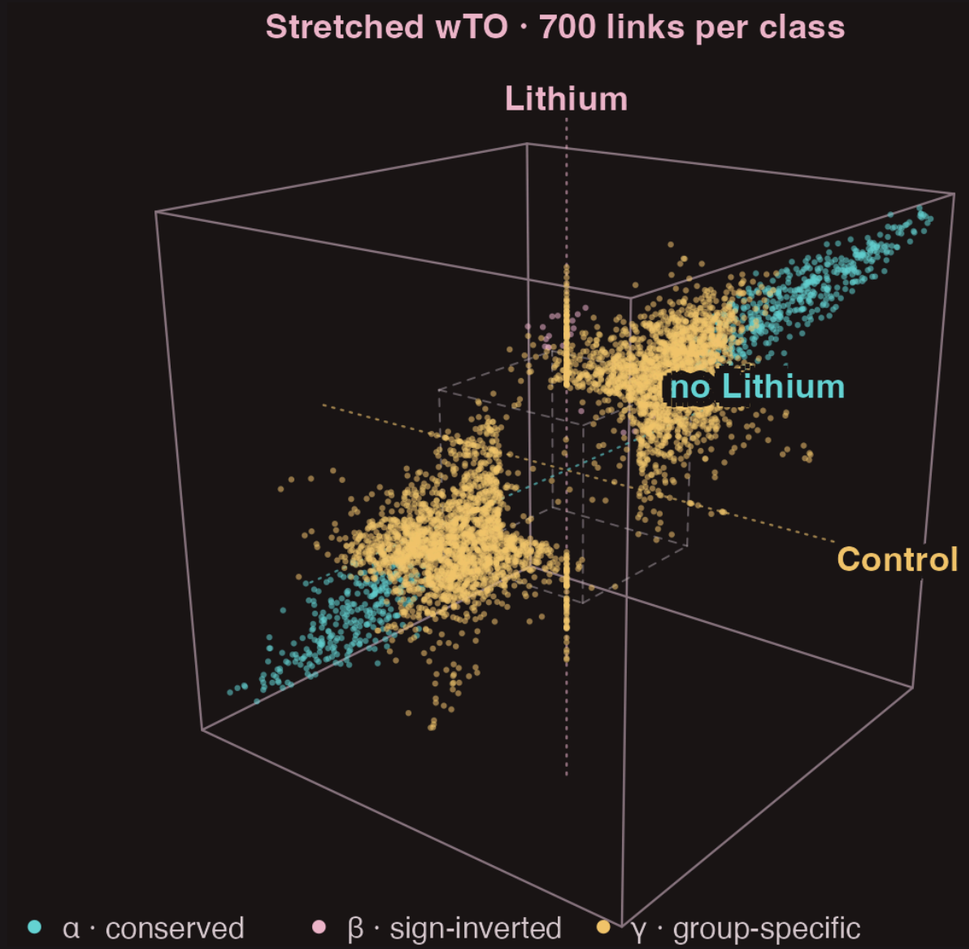
CoDiNA · Run the Differential-Network Comparison

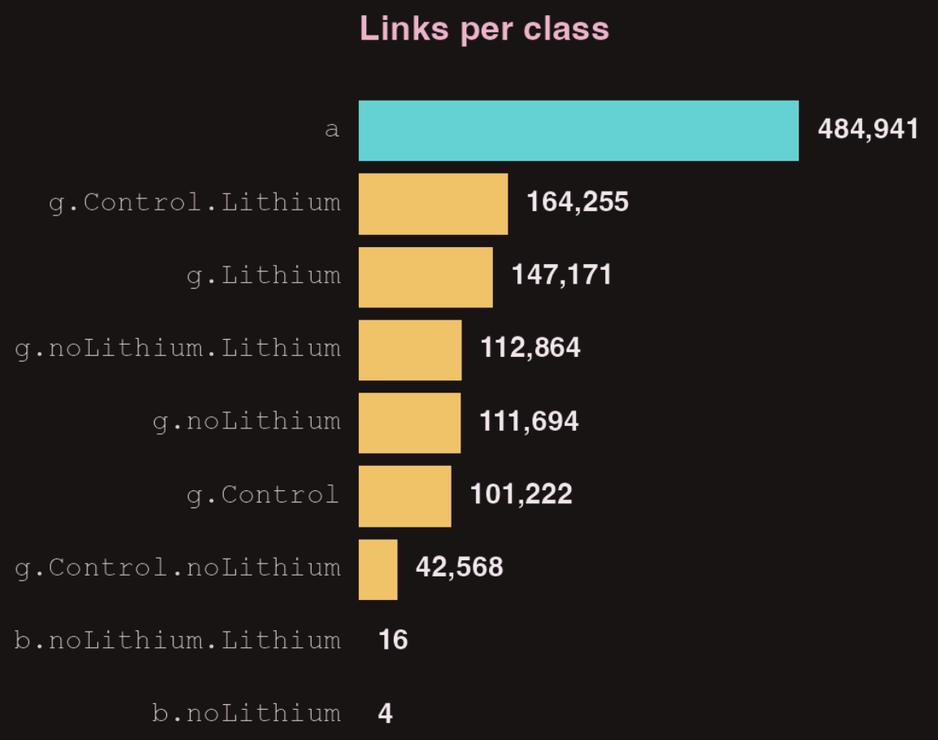
```r
to_codina_edges <- function(x, padj = 0.05) {   # keep stable links only
  x |> filter(pval.adj < padj) |> transmute(Node.1, Node.2, wTO)
}
codina_result <- CoDiNA::MakeDiffNet(
  Data = list(to_codina_edges(wto_control), to_codina_edges(wto_nolithium),
              to_codina_edges(wto_lithium)),
  Code = c("Control", "noLithium", "Lithium"), cutoff = 0.33,
  stretch = TRUE   # rescale each network by its max |wTO| (unequal group sizes)
)
sort(table(codina_result$Phi_tilde), decreasing = TRUE) |> as.data.frame() |>
  setNames(c("Phi_tilde", "links"))
```

```
           Phi_tilde  links
1                  a 484941
2   g.Control.Lithium 164255
3          g.Lithium 147171
4 g.noLithium.Lithium 112864
5        g.noLithium 111694
6          g.Control 101222
7 g.Control.noLithium  42568
8 b.noLithium.Lithium     16
9        b.noLithium      4
```

Input: the stable wTO links of each group (absent = 0). **Output:** one class per pair. **Why `stretch`:** groups of 112, 74 and 152 samples. Correlations from smaller groups are noisier and run larger in |value|, so without rescaling γ links pile up in the smallest group.

CoDiNA · Every Real Link Is a Point in the Cube





Each point is a real gene pair with its three stretched weights. Points cluster toward the corners that define their class; the bars count **every** link.

CoDiNA · From Links to Genes

```
codina_nodes <- CoDiNA::ClusterNodes(
  codina_result,
  cutoff.external = 0,   # keep links by Score_Phi_tilde
  cutoff.internal = 1    # and by Score_internal
)
gene_classes <- sort(table(codina_nodes$Phi_tilde),
                     decreasing = TRUE)
setNames(as.data.frame(gene_classes),
         c("Phi_tilde", "genes"))
```

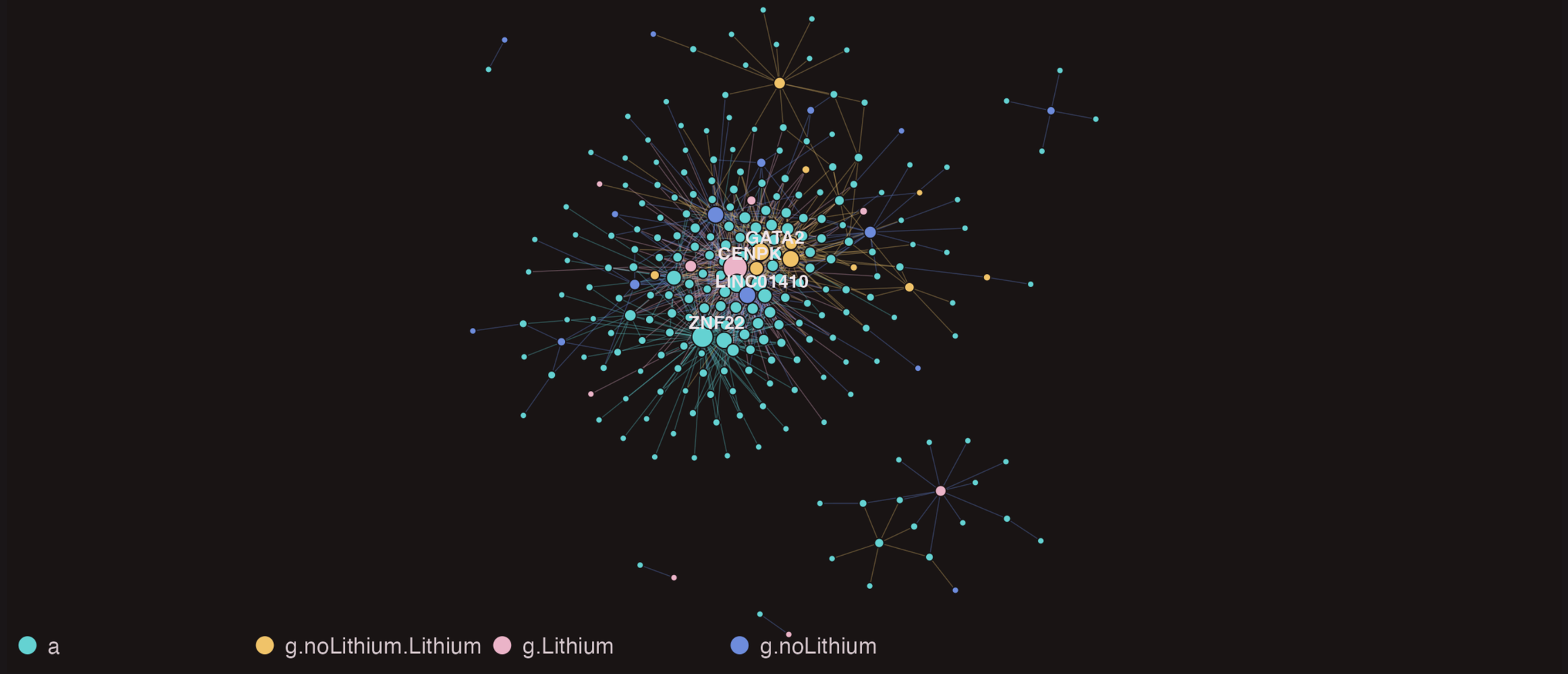
```
           Phi_tilde genes
1                  a  1396
2   g.Control.Lithium   186
3           g.Lithium   168
4           g.Control    95
5         g.noLithium    94
6 g.noLithium.Lithium    57
7 g.Control.noLithium     4
```

Each gene inherits the classes of its links. `ClusterNodes()` keeps a class only when a **chi-square test** shows the gene's links concentrate there (BH < 0.05); otherwise the gene is **U**.

γ.noLithium.Lithium: partners gained in **both** patient groups, the disease signature. **γ.Lithium**: only on lithium. **γ.Control.noLithium**: lost on lithium.

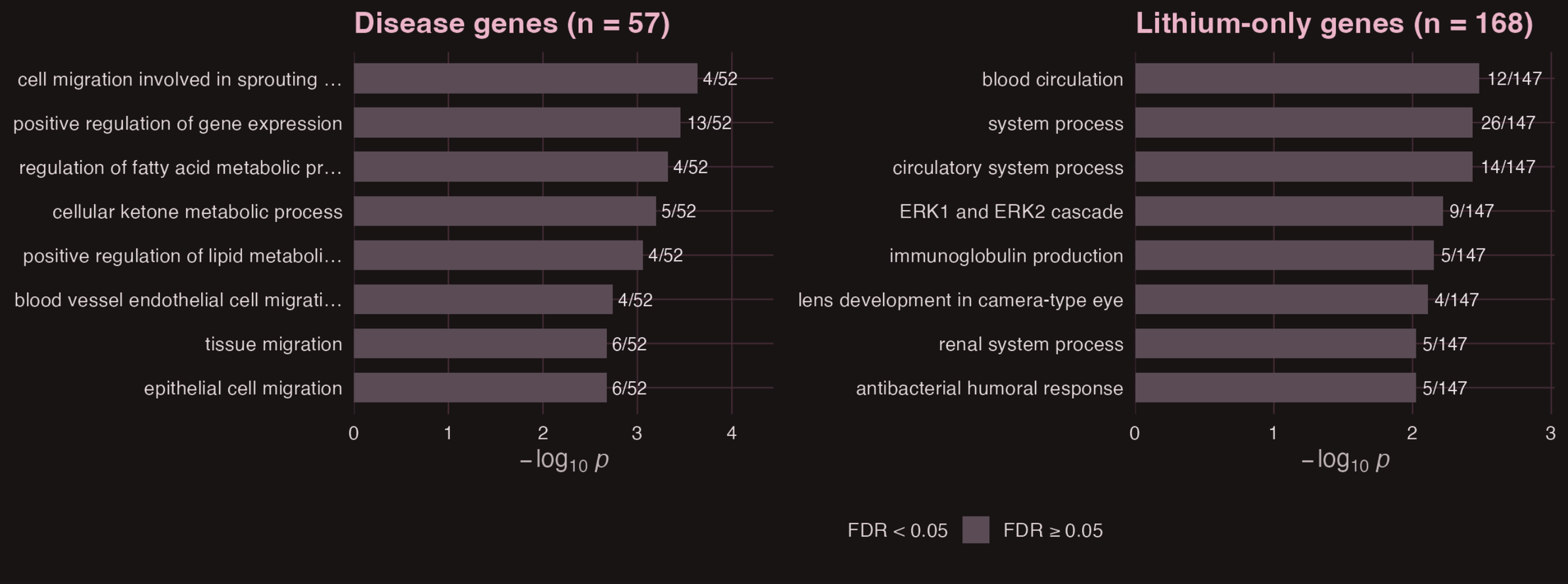


CoDiNA · The Rewired Neighbourhood



The 200 best-classified links (highest `Score_ratio`) of each of four classes. Node and link colour = class: **conserved**, **disease**, **lithium only**, and off-lithium only (blue). Labels mark the most connected gene of each class.

CoDiNA · What Do the Disease and Lithium-only Genes Do?



GO BP against the 2,000-gene universe: **0** terms pass FDR < 0.05 for the disease genes and **0** for the lithium-only genes. So we go from **what changed** (CoDiNA) to **which genes** (ClusterNodes) to **what they might do** (ORA, against the right universe).

CoDiNA · Interpreting Edge Classes

Class	Relationship across networks	Interpretation
α	Present in all networks, same sign	Conserved candidate edge
β	Present in all, sign differs in some	Sign-inverted candidate edge
γ	Present in some networks only	Group-specific candidate edge

These classes describe changes in **association**. They do not prove that a regulatory mechanism was rewired.

> Prioritize links with high `Score_ratio`, then ask whether their genes cluster into a coherent, testable program.

Hands-on · CoDiNA

Exercises 9–10 · ≈ 10 min · hands-on notebook, section 9

9. Re-run `MakeDiffNet()` with `stretch = FALSE`. How do the γ classes shift, and why does group size matter?
10. Use `ClusterNodes()` to list the **disease** genes (γ.noLithium.Lithium) and the **lithium-only** genes (γ.Lithium).

Answer · Exercise 9 · Why Stretch?

```
nets3 <- list(control = wto_control, no_lithium = wto_nolithium,
              lithium = wto_lithium)
sapply(nets3, function(x) quantile(abs(x$wTO), c(0.5, 0.9, 0.99)))
##     control no_lithium lithium
## 50%   0.138      0.166   0.145
## 90%   0.311      0.341   0.308
## 99%   0.428      0.433   0.408
codina_raw <- suppressMessages(CoDiNA::MakeDiffNet(
  Data = lapply(nets3, to_codina_edges),
  Code = c("Control", "noLithium", "Lithium"),
  cutoff = 0.33, stretch = FALSE))
lv <- sort(unique(codina_result$Phi_tilde))
cbind(stretch = table(factor(codina_result$Phi_tilde, lv)),
      raw = table(factor(codina_raw$Phi_tilde, lv)))
##                       stretch   raw
## a                      484941 90802
## b.noLithium                 4     0
## b.noLithium.Lithium        16     0
## g.Control              101222 24664
## g.Control.Lithium      164255 13639
## g.Control.noLithium     42568 22926
## g.Lithium              147171 13091
## g.noLithium            111694 87739
## g.noLithium.Lithium    112864 19994
```

The networks do not share one scale: the 74-patient network sits **highest** across its whole lwTOI distribution. On the raw scale, links cross 0.33 most often in that group, so γ links pile up in `g.noLithium` and the other classes shrink.

`stretch = TRUE` rescales each network by its own maximum lwTOI first. Group size is one reason for the different scales (fewer samples, noisier and larger lrl); real biology is another.

Answer · Exercise 10 · Disease and Lithium-only Genes

```
c(disease = length(disease_genes),
  lithium_only = length(case_genes))
##      disease lithium_only
##           57          168
head(sort(disease_genes), 8)
## [1] "ADISSP"    "ALDH2"     "ATP9A"     "BUB1"
## [5] "C14orf132" "CAPN12"    "CCN3"      "CD14"
head(sort(case_genes), 8)
## [1] "ABCG2"   "ADAM23"  "ADGRE1"  "AKAP12"  "ALOX15"
## [6] "ANGPT1"  "ANKDD1A" "ANKUB1"
```

57 disease genes (`g.noLithium.Lithium`) and **168** lithium-only genes (`g.Lithium`).

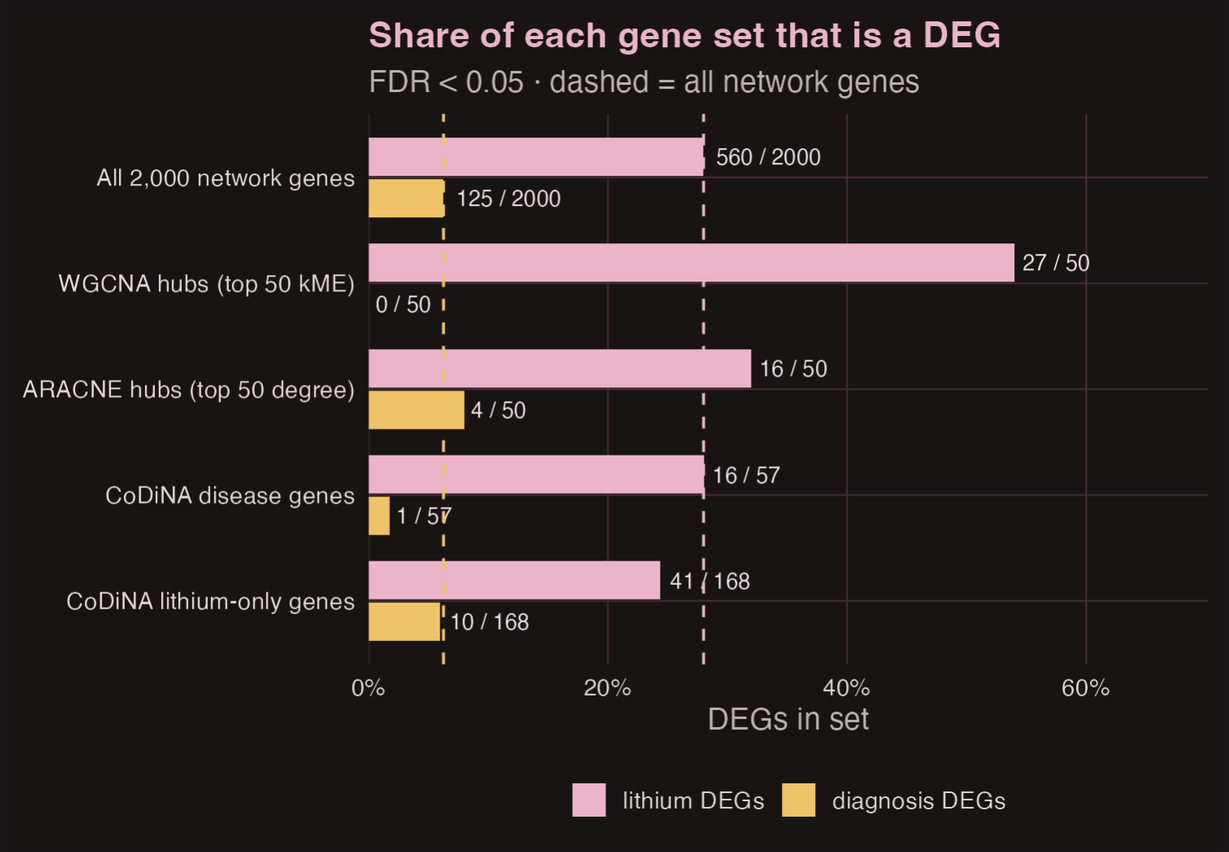
Disease genes are connected in **both** patient groups but not in controls, so lithium cannot explain them. Lithium-only genes gain partners only in treated patients. A gene is classified only when a chi-square test says its links concentrate in one class; otherwise it is **U**.

Network Change Is Not Differential Expression

Analysis	What changes?	Typical result
Differential expression (limma)	Mean expression by group	Ranked genes or DEGs
Co-expression network	Pairwise association structure	Modules and weighted edges
Differential network (CoDiNA)	Edge presence or sign across groups	Conserved or rewired candidates

A gene can have a similar average expression in all groups while its relationships with other genes differ. A gene can be a DE hit, a network candidate, both, or neither. The lists are not supposed to agree: each method sees something the others miss.

DE and Networks, Gene by Gene



limma tested all 13,013 expressed genes. **4,958** differ between patients on and off lithium (**lithium DEGs**) and **495** between patients off lithium and controls (**diagnosis DEGs**), at FDR < 0.05. Only **560** and **125** of them are among the 2,000 network genes: in blood, the most **variable** genes differ between people (HLA genotype, cell mix), not between groups.

Bars **above** the dashed baseline concentrate DE signal; bars **below** it point at genes that DE alone would miss.

Part V · From Structure to Function

ORA · GSEA

What Can We Enrich?

Gene sets from the network:

- genes assigned to a WGCNA module
- intramodular hubs (high kME)
- hubs or neighborhoods from ARACNE or wTO
- genes in CoDiNA β or γ links

Rankings from the network:

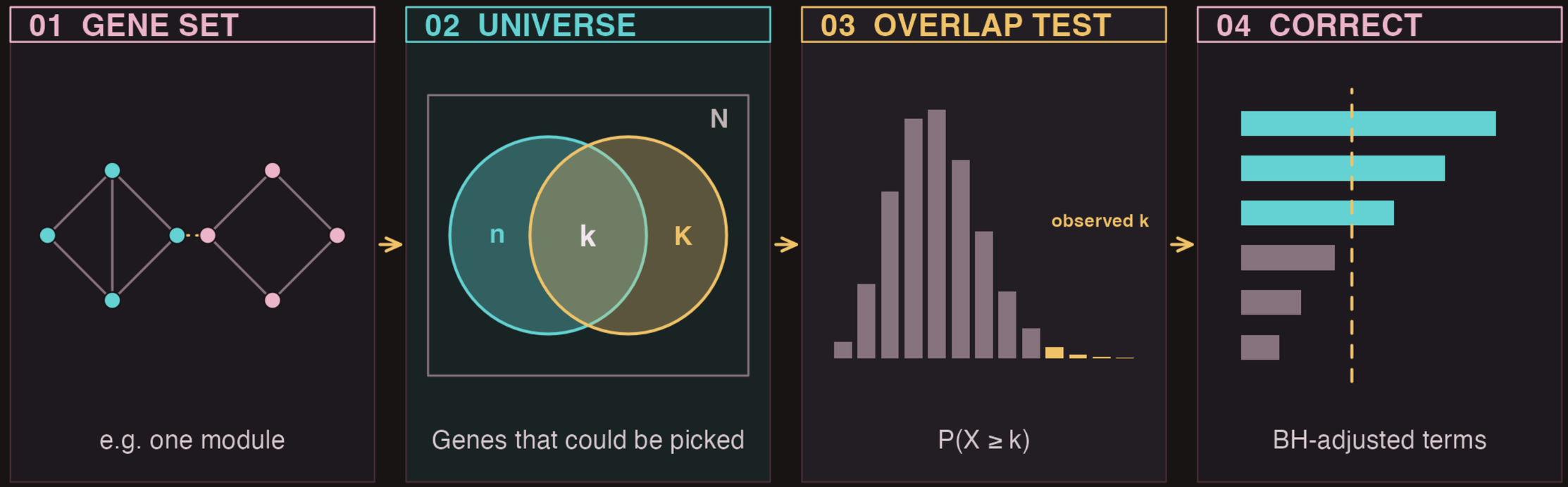
- kME for one module
- node strength or centrality

Annotation resources:

- Gene Ontology (BP, CC, MF)
- KEGG, Reactome, WikiPathways
- MSigDB hallmarks and cell-type signatures

State the resource, its version, and the **background universe**. Here the universe is the 2,000 genes that could enter the network, not the whole genome.

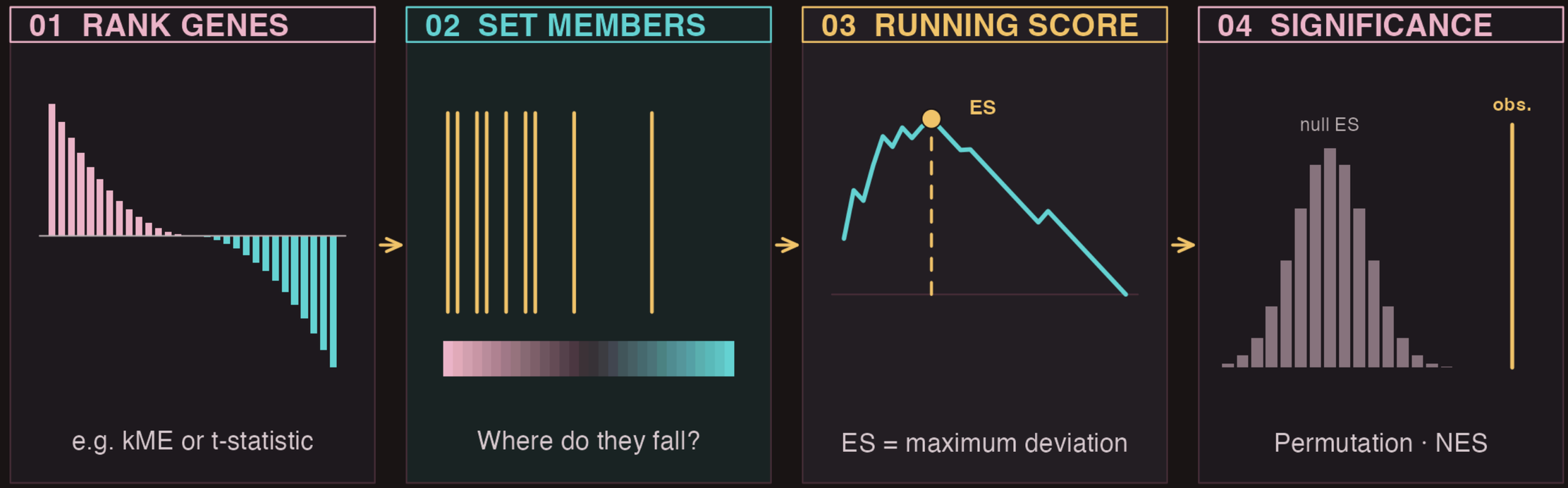
ORA · How It Works



$$p = P(X \geq k) = \sum_{x=k}^{\min(n,K)} \frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}$$

N universe genes, K in the pathway, n in the module, k in both. A large k relative to nK/N means **over-representation**. Correct the p-values across all tested terms (Benjamini–Hochberg).

GSEA · How It Works



Walk down the ranking: the score **rises** at each set member (weighted by $|r_i|$) and **falls** otherwise. The enrichment score ES is the maximum deviation from zero. Permutations give its null distribution, and NES normalizes ES for set size. **No cutoff is needed.**

ORA or GSEA?

Method	Input	Best suited to	Main caveat
ORA	A selected gene list + universe	Testing a defined module or candidate set	Depends on the list cutoff and universe
GSEA	A ranked gene list	Detecting coordinated shifts across a ranking	Depends on a defensible ranking statistic

Both require identifier mapping, an appropriate gene-set resource, and multiple-testing correction.

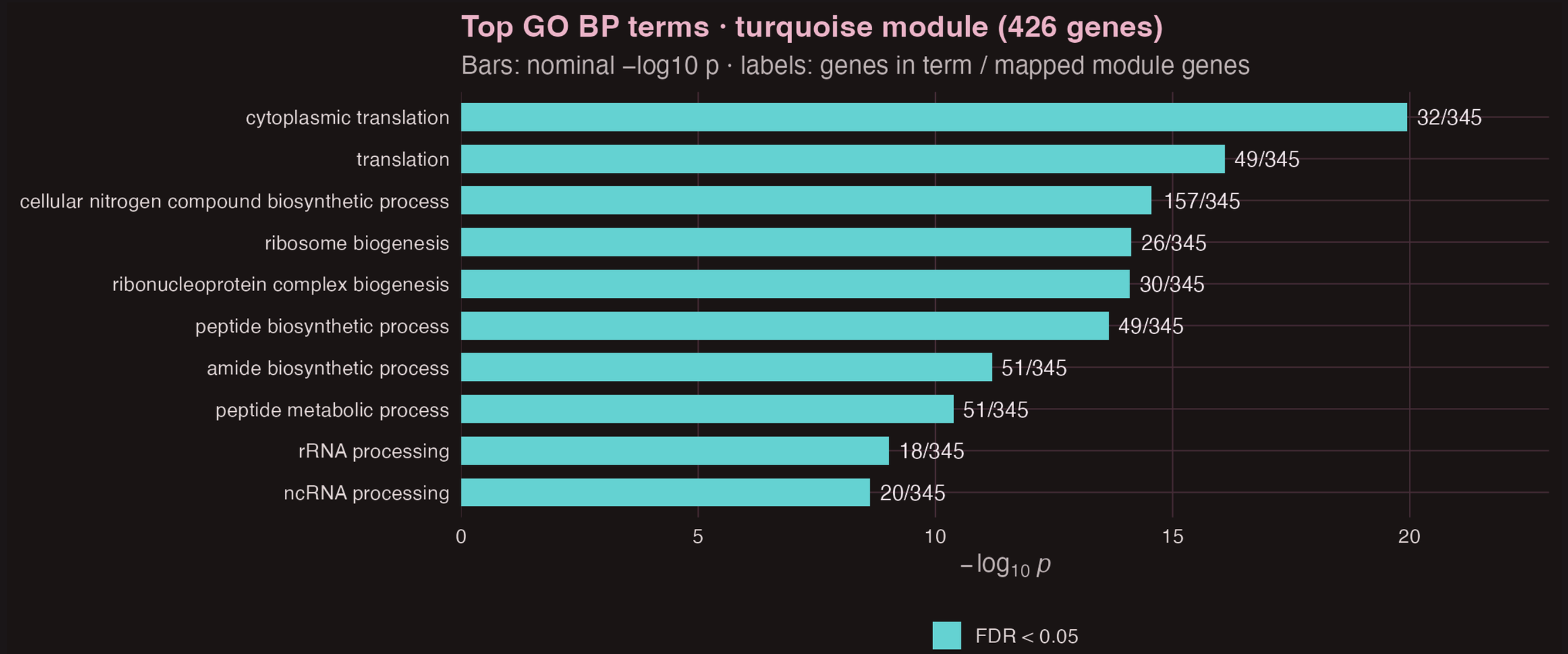
Choose ORA for a well-justified discrete set with an explicit universe. **Choose GSEA** when the full ranked statistic is meaningful. Neither method rescues a biased gene list or an inappropriate background.

Example · GO Over-representation

```r
module_sizes <- sort(table(module_colors[module_colors != "grey"]), decreasing = TRUE)
selected_module <- names(module_sizes)[1]
module_genes <- names(module_colors[module_colors == selected_module])
to_entrez <- function(symbols) clusterProfiler::bitr(
  symbols, fromType = "SYMBOL", toType = "ENTREZID", OrgDb = org.Hs.eg.db::org.Hs.eg.db)
gene_map <- to_entrez(module_genes)
background_map <- to_entrez(selected_genes)          # universe = genes that entered the network
ora_result <- clusterProfiler::enrichGO(
  gene = unique(gene_map$ENTREZID), universe = unique(background_map$ENTREZID),
  OrgDb = org.Hs.eg.db::org.Hs.eg.db, keyType = "ENTREZID", ont = "BP",
  pAdjustMethod = "BH", pvalueCutoff = 1, qvalueCutoff = 1   # keep all; filter on FDR below
)
c(module = selected_module, genes = length(module_genes),
  significant_terms = sum(as.data.frame(ora_result)$p.adjust < 0.05))
```

```
        module           genes significant_terms
  "turquoise"          "426"            "93"
```

ORA · Reading the Result



Here, **93 terms** pass FDR < 0.05 against the 2,000-gene universe. A whole-genome background can inflate enrichment for pathways that were more likely to enter the gene set in the first place. Always test against the genes that **could** have been selected.

Example · GO Gene Set Enrichment Analysis

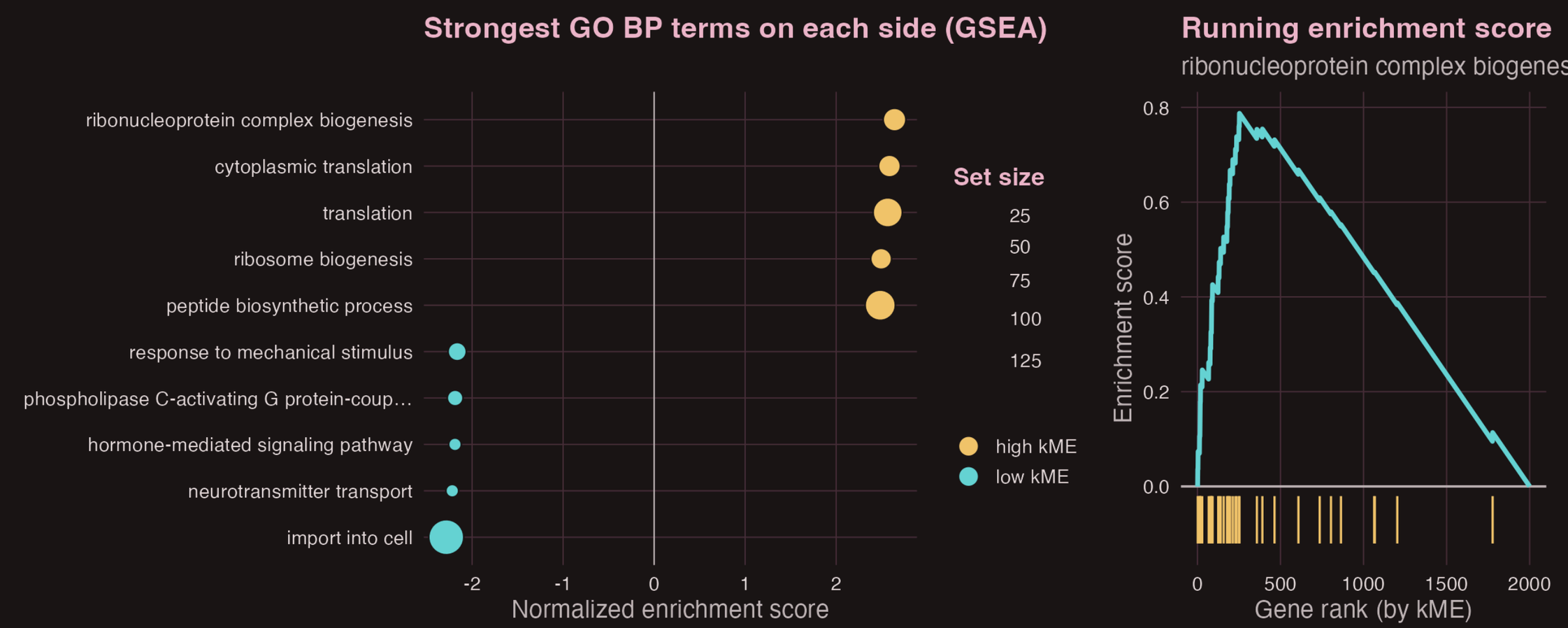
```r
kme_column <- paste0("kME", sub("^ME", "", selected_module))
gene_rank <- sort(setNames(kme_matrix[, kme_column], rownames(kme_matrix)), decreasing = TRUE)
rank_map <- to_entrez(names(gene_rank))
rank_table <- data.frame(SYMBOL = names(gene_rank), score = as.numeric(gene_rank)) |>
  inner_join(rank_map, by = "SYMBOL") |>
  distinct(ENTREZID, .keep_all = TRUE)
gene_rank_entrez <- sort(setNames(rank_table$score, rank_table$ENTREZID), decreasing = TRUE)

gsea_result <- clusterProfiler::gseGO(
  geneList = gene_rank_entrez, OrgDb = org.Hs.eg.db::org.Hs.eg.db,
  ont = "BP", keyType = "ENTREZID", pAdjustMethod = "BH",
  pvalueCutoff = 0.05, seed = TRUE
)
nrow(as.data.frame(gsea_result))
```

```
[1] 277
```

Genes are ranked by **module membership (kME)** for the turquoise module. GSEA asks which pathways concentrate among the module's most central genes, without choosing a cutoff.

GSEA · Reading the Result



Positive NES: the pathway concentrates among the module's most central genes (high kME). **Negative NES:** among the genes least connected to it. Gold ticks mark set members along the ranking.

Hands-on · Enrichment

Exercises 11–13 · ≈ 15 min · hands-on notebook, sections 10–11

11. ORA on the γ.Lithium genes with the 2,000-gene universe vs the whole genome: how many terms pass FDR in each?
12. Rank all genes by kME of the trait-associated module and run GSEA. Which terms have the largest NES?
13. Flag each network gene as DEG, WGCNA hub, ARACNE hub, or CoDiNA class. Which genes do DE and networks agree on?

Answer · Exercise 11 · The Background Matters

```
ids <- function(g) unique(to_entrez(g)$ENTREZID)
ora_bg <- function(genes, universe = NULL)
  clusterProfiler::enrichGO(
    gene = ids(genes),
    universe = if (!is.null(universe)) ids(universe),
    OrgDb = org.Hs.eg.db::org.Hs.eg.db,
    keyType = "ENTREZID", ont = "BP", pAdjustMethod = "BH",
    pvalueCutoff = 1, qvalueCutoff = 1)
n_terms <- function(x)
  if (is.null(x)) 0 else sum(as.data.frame(x)$p.adjust < 0.05)
ora_counts <- c(
  network_universe = n_terms(ora_bg(case_genes, selected_genes)),
  whole_genome = n_terms(ora_bg(case_genes)))
ora_counts
## network_universe     whole_genome
##                0                0
```

0 terms pass FDR < 0.05 against the 2,000 network genes and **0** against the whole genome.

The two runs ask different questions: are the lithium-only genes special **among variable blood genes**, or **among all annotated genes**? Terms can appear or disappear in either direction. Whichever background you choose, report it.

Answer · Exercise 12 · GSEA on Module Membership

```
rank_tbl <- tibble(SYMBOL = names(kme_top), score = kme_top) |>
  inner_join(to_entrez(names(kme_top)), by = "SYMBOL") |>
  distinct(ENTREZID, .keep_all = TRUE)
ranks <- sort(setNames(rank_tbl$score, rank_tbl$ENTREZID), decreasing = TRUE)
gsea_trait <- clusterProfiler::gseGO(
  ranks, OrgDb = org.Hs.eg.db::org.Hs.eg.db, ont = "BP",
  keyType = "ENTREZID", pvalueCutoff = 0.05, seed = TRUE, verbose = FALSE)
as.data.frame(gsea_trait) |> arrange(desc(NES)) |> head(5) |>
  transmute(term = trunc_txt(Description, 45), setSize,
            NES = round(NES, 2), padj = signif(p.adjust, 2))
##                                       term setSize  NES    padj
## GO:0009755 hormone-mediated signaling pathway      15 2.11 3.3e-03
## GO:0001659           temperature homeostasis      27 2.10 1.2e-03
## GO:0098657                 import into cell     131 2.07 3.7e-05
## GO:0030029       actin filament-based process      99 2.06 1.3e-04
## GO:1990845             adaptive thermogenesis      23 2.04 4.0e-03
```

Genes are ranked by kME for the **black** module. A positive NES means the term is concentrated among the genes most tied to the module; no cutoff was needed. Compare these terms with the hubs from Exercise 5.

Answer · Exercise 13 · Do DE and Networks Agree?

```
wgcna_hubs <- names(head(sort(kme_top[module_colors == top_color],
                              decreasing = TRUE), 50))
flags <- tibble(gene = selected_genes) |>
  mutate(DEG = gene %in% deg_genes,
         WGCNA_hub = gene %in% wgcna_hubs,
         ARACNE_hub = gene %in% aracne_hubs,
         CoDiNA_lithium = gene %in% case_genes,
         network = WGCNA_hub | ARACNE_hub | CoDiNA_lithium)
with(flags, table(DEG, network))
##        network
## DEG     FALSE TRUE
##   FALSE  1258  182
##   TRUE    479   81
c(disease_genes = length(disease_genes),
  also_diagnosis_DEG = sum(disease_genes %in% dx_genes))
##      disease_genes also_diagnosis_DEG
##                 57                  1
```

81 genes are flagged by both DE and a network method; **182** only by a network method. Of the 57 disease genes, 1 are diagnosis DEGs.

The methods answer different questions, so the lists overlap only partly. Genes found by **both** are the strongest candidates; network-only genes are where co-expression adds something DE cannot see.

Part VI · Wrap-up

Compare · reflect · reproduce

Comparing the Network Methods

Method	Edge score	Main output	Strength	Limitation
WGCNA	Correlation$^{\beta}$, TOM	Modules, eigengenes, kME	Modular organization, trait links	Keeps indirect edges; unsigned by default
ARACNE	Mutual information + DPI	Sparse network	Nonlinear; prunes indirect links	Needs many samples
wTO	Signed correlation + neighbors	Signed edges with p-values	Sign-aware, resampling-based	Resampling is costly
Consensus	Weighted mean of wTO	Robust edges	Reproducibility across cohorts	Needs comparable datasets
CoDiNA	Geometry across networks	α, β, γ classes + scores	Condition differences	Inherits input-network quality

Parameter Sensitivity and Methodological Care

Choice to perturb	What to compare
WGCNA `power`, `minModuleSize`, `mergeCutHeight`	Module assignments and phenotype associations
ARACNE estimator and `eps`	Edge weights and network density
wTO resampling method, `n`, and edge threshold	Edge stability and network density
CoDiNA `cutoff` and `stretch`	α, β, and γ classifications
Enrichment universe, database, and ID mapping	Terms, FDR, and leading genes

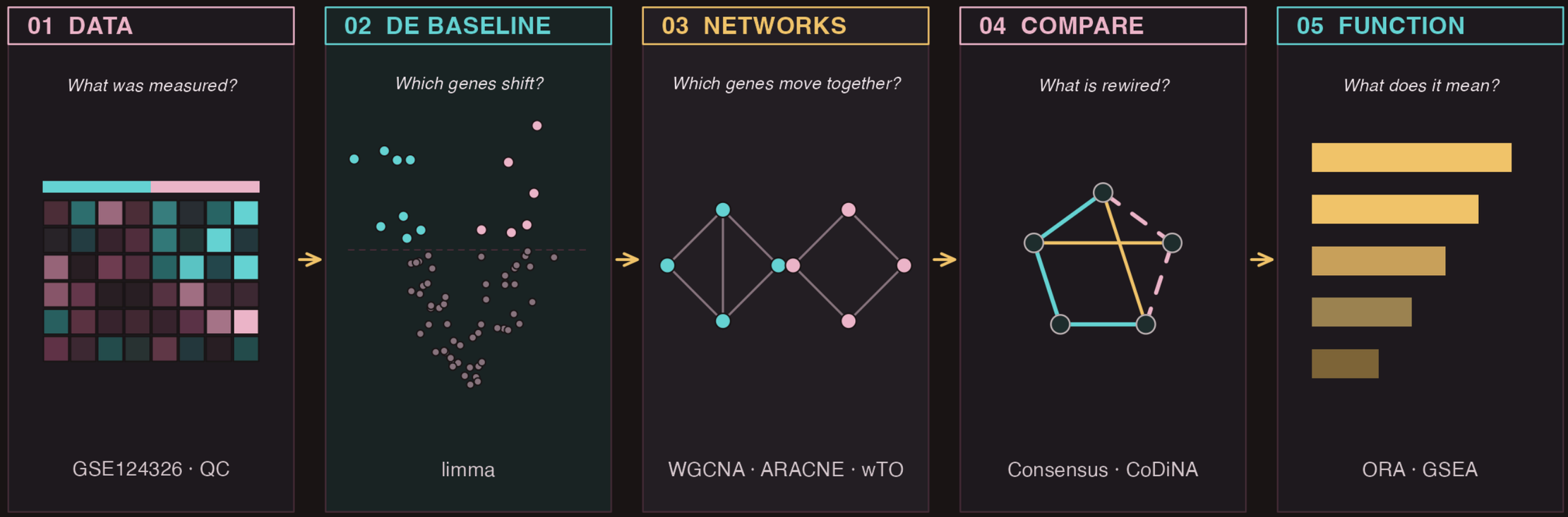
Define the background gene universe, correct for multiple testing, report unmapped IDs, and validate prioritized findings in an independent cohort when possible.

Limitations and Common Pitfalls

1. Correlation does not imply causation or physical interaction.
2. Few samples per condition make individual edges uncertain.
3. Gene-ID mapping can reduce the coverage of functional analyses.
4. Network and enrichment results depend on thresholds and databases.
5. Candidate modules and edges need independent or experimental validation.

> Everything we built today generates **hypotheses**. None of it is a finding until it holds up in an independent cohort.

The Full Route, Once More



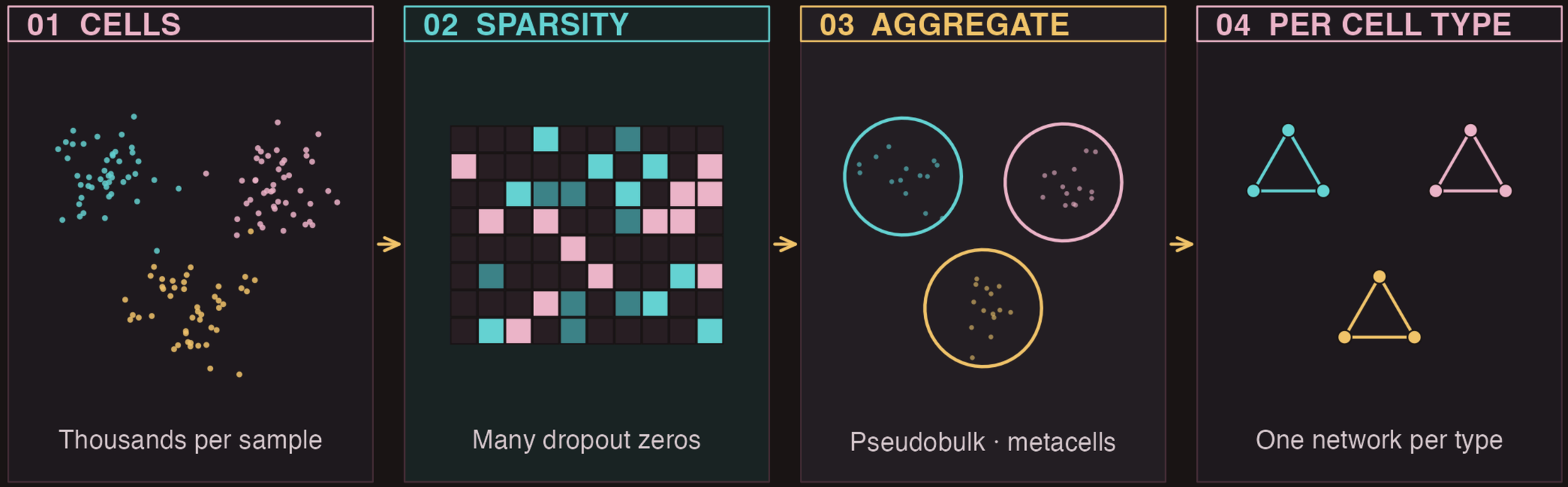
> Each stop answers a different question. Together they turn a matrix of numbers into a **short list of genes you can test**.

Conclusion

Back to lithium and bipolar disorder:

1. In blood, the loudest co-expression comes from **people, plates, and cell mix**. Adjust and filter before building anything.
2. Lithium shifts thousands of genes, each by a little, and part of its signal travels with neutrophil counts. It was not randomized, so it stays a hypothesis.
3. With three groups, CoDiNA separates links shared by **all patients** (disease) from links seen **only on lithium** (drug). Set next to the DEGs, they show what co-expression adds.
4. WGCNA, ARACNE, and wTO define a link differently; consensus asks what is **robust**, CoDiNA what **changed**, and ORA and GSEA what it might mean, given a stated universe.
5. Report every parameter, and validate before claiming mechanism.

From Bulk to Single Cells



The challenge: single-cell matrices are sparse, and correlations across cells mix **cell-type composition** with true co-regulation.

The strategy: aggregate first (**pseudobulk** per sample and cell type, or **metacells** as in hdWGCNA), then apply today's methods **within** each cell type.

Where to Go Next

Go deeper

- **WGCNA tutorials** · consensus modules across datasets, module preservation
- **wTO** and **CoDiNA** vignettes · more than two networks, interactive plots
- **hdWGCNA** · co-expression for single-cell and spatial data
- **SCENIC** · regulatory networks from single cells

Share and explore

- Export to **Cytoscape**: `igraph::write_graph(g, "net.graphml", "graphml")` or `RCy3`
- Network medicine workshop: https://deisygysi.github.io/NetMed_Workshop/
- Today's hands-on notebook and the cached networks in `results/`

> Start from a question, choose the definition of a link that matches it, and report every choice.

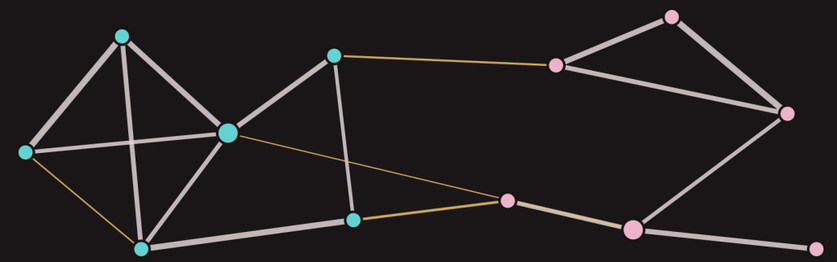


References

- Aibar, S. et al. (2017). SCENIC: single-cell regulatory network inference and clustering. *Nat Methods*, 14, 1083–1086.
- Barabási, A.-L. & Albert, R. (1999). Emergence of scaling in random networks. *Science*, 286, 509–512.
- Barabási, A.-L. & Oltvai, Z. N. (2004). Network biology: understanding the cell's functional organization. *Nat Rev Genet*, 5, 101–113.
- Gysi, D. M. et al. (2018). wTO: an R package for computing weighted topological overlap and consensus networks. *BMC Bioinformatics*, 19, 392.
- Gysi, D. M. et al. (2020). Whole transcriptomic network analysis using Co-expression Differential Network Analysis (CoDiNA). *PLOS ONE*, 15(10), e0240523.
- Krebs, C. E. et al. (2020). Whole blood transcriptome analysis in bipolar disorder reveals strong lithium effect. *Psychol Med*, 50, 2575–2586.
- Langfelder, P. & Horvath, S. (2008). WGCNA: an R package for weighted correlation network analysis. *BMC Bioinformatics*, 9, 559.
- Margolin, A. A. et al. (2006). ARACNE: an algorithm for the reconstruction of gene regulatory networks. *BMC Bioinformatics*, 7(Suppl 1), S7.
- Meyer, P. E., Lafitte, F. & Bontempi, G. (2008). minet: an R/Bioconductor package for inferring large transcriptional networks using mutual information. *BMC Bioinformatics*, 9, 461.
- Morabito, S. et al. (2023). hdWGCNA identifies co-expression networks in high-dimensional transcriptomics data. *Cell Rep Methods*, 3, 100498.
- Newman, M. E. J. (2006). Modularity and community structure in networks. *PNAS*, 103, 8577–8582.
- Nowick, K. et al. (2009). Differences in human and chimpanzee gene expression patterns define an evolving network of transcription factors in brain. *PNAS*, 106, 22358–22363.
- Smyth, G. K. (2004). Linear models and empirical Bayes methods for assessing differential expression in microarray experiments. *Stat Appl Genet Mol Biol*, 3, Article 3.
- Subramanian, A. et al. (2005). Gene set enrichment analysis. *PNAS*, 102, 15545–15550.
- Yu, G. et al. (2012). clusterProfiler: an R package for comparing biological themes among gene clusters. *OMICS*, 16(5), 284–287.
- Zhang, B. & Horvath, S. (2005). A general framework for weighted gene co-expression network analysis. *Stat Appl Genet Mol Biol*, 4, Article 17.

Thank You

Co-expression Networks: Construction and Analysis



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